

Program for International Student Assessment (PISA 29)

I. INTRODUCTION

The National Center for Education Statistics (NCES) within the Institute of Education Sciences (IES) of the U.S. Department of Education (ED) has the following mission:

1. to collect and analyze education information and statistics in a manner that meets the highest methodological standards;
 2. to report education information and statistics in a timely manner; and
 3. to collect, analyze, and report education information and statistics in a manner that:
 - a. is objective, secular, neutral, and non-ideological and is free of partisan political influence and racial, cultural, gender, or regional bias; and
 - b. is relevant and useful to practitioners, researchers, policymakers, and the public.
- P.L. 107-279, Part C, Sec. 151(b); H.R. 3801, Part C, Sec. 151(b), 2002.

NCES participates in international student assessments in order to provide comparable indicators on student performance and schooling practices across countries in order to benchmark U.S. student performance. To fulfill this purpose, the United States must follow through with well-organized and well-executed data gathering activities within our national boundaries. These efforts will allow NCES to provide policy-relevant data for comparisons of the U.S. education system with education systems in other countries not only related to characteristics of the education system, but also as the outcome of the education process related to participation in further education and training, readiness for the work force, and participation in society more generally. High quality data of this kind will help to inform decision making on the part of national, state, and local policy makers.

A. Purpose of this Procurement

NCES has a requirement to participate in and gather data from The Program for International Student Assessment (PISA). Details on the purpose, conduct, and the international sponsoring organization (OECD) follow.

B. Legislative Authority

The authority for these studies can be found in NCES' longstanding mission to "collect, and analyze and disseminate statistics and other information related to education in the United States and in other nations" (stated in section 406 of the General Education Provisions Act, as amended (20 U.S.C. 1221e-1) and in the Education Sciences Reform Act of 2002 (P.L. 279-107, Part C, Sec. 153):

- (a) GENERAL DUTIES.-The Statistics Center shall collect, report, analyze, and disseminate statistical data related to education in the United States and in other nations, including...(6) acquiring and disseminating data on educational activities and student achievement (such as the Third International Math Study) in the United States compared with foreign nations.

(See <http://www.ed.gov/policy/rschstat/leg/PL107-279.pdf> for the full Education Sciences Reform Act.)

C. Quality Control

NCES is firmly committed to determining how well each project meets 1) the project goals, 2) the quality of data collected, and 3) implications for future projects. Project quality control should be integrated into every phase of the project. For instance, potential problems should be identified early so that possible solutions are tested as part of the field test. Item reliability and validity tests for questionnaires and student cognitive tests must be built into the data collection efforts, and results from these tests shall inform data collection and processing. Nonresponse analyses shall be conducted as part of the final weighting process, and survey methodology and sample design effects shall be documented for use by all data users. Thus, as in other NCES projects, statistical quality control shall be incorporated into each stage of this project as appropriate.

Quality control also includes (1) managing staff in the field to ensure the high quality of their performance; (2) instituting procedures relevant to confidentiality and protection of data (including the secure transmittal of confidential data); and (3) ensuring deliverables are submitted on time and are of sufficient quality (in terms of accuracy, coherence, clarity, and comprehensiveness) to meet NCES statistical standards, OECD/PISA standards, professional standards, and NCES and IES style guidelines.

D. NCES Standards

NCES standards set guidelines to ensure the quality of NCES surveys, data, data products, and statistics. All work conducted under this contract must, at minimum, meet the standards and guidelines set forth in the most currently available statistical standards and guidelines for the National Center for Education Statistics (NCES) and as amended by standards implemented since that time. The NCES Standards can be accessed at <https://nces.ed.gov/statprog/2012/>. Publication style guidelines for IES products shall also be followed. The guidelines can be accessed at <https://nces.ed.gov/statprog/style.asp>.

All final deliverables that are to be assigned a NCES publication number or that will be made public through the NCES website as a product will be delivered in PDF format (or another appropriate format approved by NCES) in accordance with the NCES web publishing requirements (available at https://nces.ed.gov/pubs2003/secureweb/ch_2.asp), and must adhere to Section 508 requirements for accessibility. All final deliverables that are to be made public through the NCES website as webpages or in a web-based reporting format must also adhere to Section 508 requirements for accessibility. More information on accessibility is available through the Federal IT Accessibility Initiative at: <http://www.section508.gov>.

E. Confidentiality

Appendix B provides a guide to issues of confidentiality.

F. Contract Type

The contract type is Firm Fixed Price.

G. Period of Performance

The anticipated total period of performance for this contract is 48 months.

- Base Period : 12 months

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- Option Periods: Additional three 12-month periods.

The estimated years within the period of performance covered by each task and subtask in the PWS are shown in Exhibit 1.

Exhibit 1. Estimated Years Covering the Period of Performance for All Tasks

Study or Activity	Years			
	1	2	3	4
Task 1: Project management				
1.1: Post-Award Project Meeting	X			
1.2: Ongoing project management	X	X	X	X
1.3: Data Security Plan and Staff Security Requirements	X	X	X	X
Task 2: Obtaining Office of Management and Budget Clearances	X		X	
Task 3: International Meetings and Communications	X	X	X	X
Task 4: Adapting, preparing, testing and producing instruments Collection				
4.1 Assessment instruments for the field test	X	X		
4.2 Assessment instruments for the main study		X	X	
Task 5: Sampling				
5.1 Sampling for the field test	X			
5.2 Sampling for the main study			X	
Task 6: Data collection				
6.1 Field trial data collection	X	X		
6.2 Main Study data collection		X	X	
6.3 Main Study data processing and disclosure risk review			X	
Task 7: Human scoring of cognitive items and coding of responses to questionnaire items				
7.1 Human scoring of the field test cognitive items		X		
7.2 Scoring of the main study cognitive items			X	
7.3 Coding responses to fill-in item in questionnaires			X	
Task 8: Teacher Questionnaires		X	X	
Task 9. Main study electronic data files and user's guides				X
Task 10. Teacher Questionnaire electronic data files and user's guides				X

II. SCOPE

Independently and not as an agent of the United States Government, the contractor shall provide all personnel, materials, services, and facilities necessary for the various studies and perform the tasks as described below.

The purpose of this contract is to implement the 2029 cycle of U.S. participation in PISA. The overall goal for the procurement is data collection and reporting in the United States that meets all international and NCES guidelines and that makes available on a timely basis high-quality internationally comparative survey and assessment data.

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The Scope of Work for the contract also includes two optional tasks, which may alter or add to the work specified above.

The contractor shall be clear that NCES is the sole authority to manage the study in the United States and to negotiate with the international consortium on any issues related to the study. The contractor shall not use the items for any other purpose without NCES' written consent. Under no circumstances shall the contractor conduct any activities related to PISA including conducting the assessment in sub-national jurisdictions (e.g., states or school districts), without permission from NCES.

H. Requirements

This section provides a description of the contract requirements (tasks) and a list of deliverables associated with each task. References herein to “the project” refer to the 2029 PISA study.

A preliminary timeline for the 2029 PISA study, with major events and activities, is in Table 1 below. The PISA 2029 Tentative Schedule and Background Information for National Preparations, provided by the OECD, is in Appendix A. Note that the U.S. will participate in the core domains of Mathematics, Reading and Science, as well as the innovative domain (Media and Artificial Intelligence Literacy) in 2029. In addition, the U.S. plans to participate in the following national options: Financial Literacy assessment and the Teacher Questionnaire.

The contractor shall report all activities for this contract and activities related to this study to (a) the Contracting Officer's Representative (COR), who will be responsible for assisting the Contracting Officer (CO) oversee the entire project.

All data, publications, and products of the study are the sole property of the U.S. government, and shall be provided to it as specified below and upon request. Under no circumstances will the contractor, or any subcontractors, distribute or release any data, publications, or products of the study in any form to any individual or party without prior permission of the COR.

The contractor shall work through the COR in all relationships and discussions with the OECD, international contractors, foreign governments, foreign education professionals, and all other agencies and individuals related to the study. The contractor shall not convey to any of these entities information that may be construed as NCES, IES, or Department of Education policy, except where related to technical aspects of the study or where given explicit permission by the COR.

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Task 1. Project management

Under this task, the contractor shall support NCES in establishing the PISA 2029 US National Center (NC) and provide project management to complete all tasks within this procurement. At the direction of the National Project Manager (NPM) for the study, the NC will act as the liaison to the OECD and carry out all national tasks required for successful implementation of the study. The NPM is the project leader for PISA at NCES. Table 1 below shows the major activities for PISA 2029.

Table 1. Schedule of major activities for PISA 2029

All dates and activities are tentative. The final schedule will be confirmed in 2026 as part of the NPM manual.

Phase	Period (approximate)	Tasks for PISA National Centres
Development phase	Jan-26 to Dec-26	Establish PISA National Centre and appoint National Project Manager Participate in item development workshop (optional) Contribute ideas for new test items (optional) Review new test items
Field Trial preparation	Jan-27 to Jan-28	Translate and adapt test instruments and manuals Review and test national version of Field Trial instruments Sampling and recruitment of schools for the Field Trial Recruitment and training of test administrators Coder training
Field Trial data collection <i>Data collection period: 1 February to 15 September 2028</i>	Feb-28 to Sep-28	Support schools to prepare and plan for testing Co-ordinate activities of test administrators Distribute materials to schools Organise and monitor data collection Coding of responses Data management and data submission
Field Trial review phase	Aug-28 to Oct-28	Review Field Trial data analysis reports Review Field Trial materials and request changes to errors Review operational procedures
Main Survey material preparation	Jun-28 to Sep-28	Main Survey - Translate and adapt test instruments
Main Survey operational preparation	Oct-28 to Jan-29	Preparation of manuals and school-level materials Review and test national version of Main Survey instruments Sampling and recruitment of schools for the Main Survey Recruitment and training of test administrators Coder training
Main Survey data collection <i>Data collection period: 1 February to 14 September 2029</i>	Feb-29 to Sep-29	Support schools to prepare and plan for testing Co-ordinate activities of test administrators Distribute materials to schools Organise and monitor data collection Coding of responses Data management and data submission
Report preparation phase	Sep-29 to Aug-30	Review national databases Review international databases Analyse national data and prepare national report Prepare dissemination of results

Subtask 1.1 Post-Award Project Meeting

Within the first week after contract award, the contractor shall hold a meeting with the COR and other appropriate NCES staff to review the overall contract tasks, and to refine the proposed plans for Project management, Staffing and succession, Data security, Invoicing, and the master project schedule. These refinements are not to alter the specifications of the contract, but to provide management information for use by both the contractor and the government in monitoring the work to be performed. This meeting will also provide the opportunity for ED staff to share project experience and materials with the contractor.

Deliverable:

1.1.1 Minutes and attendee list from post-award meeting focusing on action items and key decisions

1.2 Ongoing project management

Within the first two weeks after contract award, the contractor shall hold a meeting with NCES staff to review the overall contract tasks. The Contractor shall also meet with the NPM, on average, every other week focusing on major decisions and action items.

The Contractor shall deliver a master schedule of project activities within one month of the initiation of the project as well as quarterly study progress and financial reports describing the activities completed during the previous month, addressing any challenges, and explaining what measures have been implemented to resolve them.

Deliverables:

- 1.2.1 Post-award project planning meeting agenda and materials
- 1.2.2 Monthly IT report
- 1.2.3 Master schedule
- 1.2.4 Quarterly study progress and financial report
- 1.2.5 Agenda for and brief notes following biweekly meetings

1.3 Data Security Plan and Staff Security Requirements

The contractor shall prepare and submit a data security plan submitted to the COR no later than 2 weeks after the start of the contract. See Appendix B for a detailed discussion of the data security requirements of the contract, including the required monthly deliverables under “Implementation of Data Security Plan”(B.4j). The contractor shall ensure that all contractor-led operations, including those of subcontractors, adhere to federal laws and standards in this regard.

In addition, all contractor staff working on the project for 30 days or more must complete a series of security forms and undergo a federal background check per OMB requirements, with higher levels of security clearance required for those who visit schools or who may come in contact with confidential data. The contractor shall ensure that all staff who may come in contact with confidential data shall

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- (a) be trained in the proper handling of confidential data, and
- (b) first sign and then notarize an affidavit of nondisclosure before having access to any data with confidential information.

The contractor shall execute these affidavits of nondisclosure, retain them in a secure location and provide a list to the COR of all the individuals who have signed affidavits within 4 weeks of the award of the contract. As new persons are assigned to the project, the security forms shall be filled out and an affidavit shall be executed for them on the first working day of assignment to the study and an updated list provided to the COR, no less than on a monthly basis. The contractor shall notify the COR if any contractor staff leave the project. The contractor shall be aware of and comply with all contractor personnel security regulations and all Department IT security policy requirements, as described in Appendix B.

Deliverables:

- 1.3.1 Draft data security plan
- 1.3.2 Final data security plan
- 1.3.3 List of all personnel with signed Affidavits of Nondisclosure

Task 2. Obtaining Office of Management and Budget Clearances

The contractor shall prepare a clearance package for obtaining Office of Management and Budget (OMB) approval of the study's field trial and main study; all non-cognitive survey instruments to be used in the study; and any other materials that (a) will be distributed to participants (e.g., schools, students, and teachers), and (b) will be used to promote or endorse the study (e.g., informational letters to state officials, endorsement request letters) in accordance with current OMB, ED, and NCES requirements for such packages.

Failure to submit the OMB clearance package by the deadline may result in the contractor being unable to collect data in a timely manner.

The contractor shall prepare a draft OMB clearance package for the field trial and the main study data collection activities no later than 9 months prior to the field trial data collection, and preferably earlier. The clearance package approval process, described at <https://digital.gov/guides/prs/prs-approval-process#content-start>, is expected to involve the following series of steps:

- Contractor shall provide a draft package to the COR (based on the last OMB-approved equivalent package for the study incorporating relevant revisions from other recent NCES study clearance packages) and shall revise it within 5 business days of receiving the COR's comments (2.3.1.1).
- The COR provides the package to the NCES OMB liaison, and the contractor shall revise the package according to comments received within 5 business days.
- The NCES OMB liaison provides the package to the ED information collection clearance office, which oversees ED's internal review, and the contractor shall revise the package within 5 business days according to comments received.

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- Substantive public comments may be received during the first (60-day) public comment period, in which case the contractor shall provide draft responses to the public comment and any needed revisions to the package materials within 5 business days.
- ED publishes a second (30-day) public comment period notice and submits the package to OMB. Substantive public comments may be received during the second public comment period, in which case the contractor shall provide draft responses to the public comments and any needed revisions to the package materials within 5 business days. The contractor shall also answer any OMB questions within 5 business days and revise the package according to final comments received within 3 business days
- In the event that the any substantive changes to any non-cognitive instruments or study procedures are introduced after the second comment period has begun, the contractor shall provide revised package materials for posting for an additional 30-day comment period and subject to OMB's review and approval.

The contractor staff shall be prepared to join the COR in person or via conference call for a review of the materials with ED and OMB staff. Note that the field trial clearance package may include main study preliminary activity and recruitment materials (the inclusion of which will affect the timing of the main study clearance package).

The contractor shall ensure that authorization by OMB has been obtained before any contact is made and data collection begins. All data collection forms will bear the approval number and expiration date assigned by OMB.

Following approval, NCES will provide the contractor with an electronic copy of OMB's official approval for the study and the final version of the approved clearance package documents. The contractor must use the final OMB-approved versions of study procedures and materials and is responsible to produce a copy of the official approval for the study upon request.

An amendment to the OMB package might be needed if changes are needed for the main study instruments and procedures based on the field trial results. If such changes are needed, the Contractor shall prepare a change memo describing any revisions made in instruments or procedures from the field trial to the main study.

Deliverables:

- 2.1 Draft OMB clearance package;
- 2.2 Revised OMB clearance package to be published to the Federal Register for comment (60-day notice)
- 2.3 Revised OMB clearance package following public comments from 60-day notice and Draft responses to the public comments
- 2.4 Responses to OMB questions on package published to the Federal Register for public comment (30-day notice)

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Task 3. International Meetings and Communications)

Under this task, the contractor shall

- maintain ongoing communication with the international contractors (Table 2), and
- participate in NPM meetings (Table 3).

Table 2. OECD contractors for the three core components

Timeline (approximate)	Meeting and format
Core A: Main contractor for the development, implementation and co-ordination	ACER (Australian Council for Educational Research) in a consortium with cApStAn and HallStat
Core B: Innovative domain development	ACER (<i>Australian Council for Educational Research</i>)
Core C: Sampling	IEA (<i>International Association for the Evaluation of Educational Achievement</i>)

Table 3. Tentative schedule of NPM and training meetings

Timeline (approximate)	Meeting and format
March 2027	<i>NPM Meeting 1 (in person)</i>
September/October 2027	<i>NPM Meeting 2 (in person)</i>
September/October 2028	<i>NPM Meeting 3 (in person)</i>

Note that the OECD is expected to provide participating countries a document titled “Working with the ACER consortium” in May of 2026 that will outline the Core A contractors’ organizational structure and roles and specify the consortium’s intended working relationships with NPMs and the national centers. The contractor should assume one staff member to attend each of the three in-person meetings listed in Table 3. Within one week of the conclusion of each meeting, the contractor shall submit a brief NPM meeting trip reports highlighting decisions and action items.

Deliverables

3.1 Brief NPM meeting trip reports highlighting decisions and action items (up to 4 pages)

Task 4. Adapting, preparing, testing and producing instruments

International versions of study instruments (typically, noncognitive “background” questionnaires and cognitive questions or “items”) will be developed by international contractors and will be provided to the contractor by the NCES. The contractor shall adapt these instruments for use in the United States (e.g., by converting British-English terms to American-English or metric measures to imperial measures). NCES regularly adds additional items of national interest to the questionnaires, such as questions related to student race/ethnicity and participation in the federal free or reduced-price lunch program. The contractor shall incorporate all such national options, as approved by the NPM for the background questionnaires.

All the instruments to be used in each study’s field test and main study must be reviewed and revised at several levels before they can be reproduced (cloned or printed) and distributed for use

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by respondents. The contractor shall incorporate into its production schedule all of the following steps for review and revision of the study instruments for each study prior to the production of the study instruments. (The levels of review indicated here are subject to change.)

- Contractor revises instruments with national adaptations and national options.
- All national adaptations and national options to instruments are submitted to OECD for review and approval.
- Contractor revises instruments based on feedback obtained in steps 3 until instruments are approved by OECD and NCES.
- The OECD-approved, nationally adapted versions of non-cognitive instruments (questionnaires) are submitted to OMB for review and comment.
- Contractor revises instruments based on feedback from OMB. Revisions suggested by OMB may require the contractor to resubmit study instruments to the OECD for re-approval. [Note: OMB has rarely required changes to study instruments.]
- Contractor finalizes nationally adapted instruments for production.
- Contractor tests all final versions of instruments in their digital-based formats for usability, reliability, and security on the computer equipment or online server that will be used for administering the study in the United States.
- Assembled instruments undergo final optical check by NCES and, if required, by the OECD
- Instruments approved and finalized for production.

The contractor is responsible for (a) reviewing the forms provided by the international contractor and ensuring they are complete and correct (e.g., reflect all agreed-upon adaptations, include all items in the correct place), and (b) any mistakes in study instruments made after final approval for production, including costs associated with the correcting the problem to the satisfaction of NCES.

All online data collection must be arranged by the contractor with the IES website team, which also has specific requirements for apps to be hosted on the NCES website. Under no circumstances may online data collection be performed by the contractor using servers that do not meet IES security standards.

Both before each field test and before each main study, the contractor shall test all instruments in their digital-based formats for usability, reliability, and security on the computer equipment or online server that will be used for administering the study in the United States.

The contractor shall be aware that final non-cognitive instruments must display the OMB approval number, expiration date, and federal paperwork reduction act notice, a final copy of which is submitted to OMB for filing.

The contractor shall be familiar with the requirements of the Government Printing Office for reproduction; all reproductions shall be done in accordance with government printing regulations <https://www.govinfo.gov/content/pkg/GPO-CPUB-101spub9/pdf/GPO-CPUB-101spub9.pdf> and with international guidelines.

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The contractor shall protect the proprietary rights of all copyrighted instruments used in the study. Individual instrument data and individual item data shall not be released or shared without the permission of NCES/ED.

The contractor shall adhere to and implement NCES data security and confidentiality procedures that may affect the production, distribution, and processing of all survey instruments. The contractor must review the important detailed information on these and other topics included in Appendix B and C.

Under this task, the contractor shall

- review field test and main study assessment instruments and questionnaires in the online deployment system¹.
- Make necessary adaptations to the assessment instruments and questionnaires following protocols provided by OECD and/or NCES.

Note that the assessment instruments and student **and teacher** questionnaires are developed in English, only minor adaptations or revisions are expected for these. Adapting and preparing the instruments involves both national options (country-specific questions added to background questionnaires such as race/ethnicity and participation in the federal free or reduced-price lunch program) and adaptations to the international versions of study instruments. Once approved by NCES, the contractor shall not alter the instruments without approval from the COR.

4.1 Assessment instruments for the field test

Under this subtask, the contractor shall review field test assessment instruments and student **and teacher** questionnaires in the online deployment system and make necessary adaptations to the assessment instruments and questionnaires.

Deliverables

- 4.1.1 Proofs or final electronic versions of field test instruments (both cognitive and non-cognitive) for submission to OECD upon NCES approval

4.2 Assessment instruments for the main study

Under this subtask, the contractor shall review field test assessment instruments and student **and teacher** questionnaires in the online deployment system and make necessary adaptations to the assessment instruments and questionnaires.

Deliverables

- 4.2.1 Proofs or final electronic versions of the main study instruments (both cognitive and non-cognitive) for submission to OECD upon NCES approval

¹ For a description of the technical aspects of the online assessment system, please see <https://schoolsupport.acer.org/hc/en-au/articles/20296676534169-Technical-requirements#01H8VC4YGMV3R5RZRJ9CXQ1SFY>

Task 5. Sampling

Under this task, the contractor shall manage all sampling-related activities for the field test and main study based on the proposed approach and methodology in its accepted offer in accordance with the PISA Technical Standards.

5.1 Sampling for the field test

Under this task, the contractor shall prepare a field study sampling plan and draw a selects a convenience sample. Although not a probability sample, the sample should represent students with varying degrees of achievement in target constructs to the degree possible. As a reference, for the 2025 field trial a sample of 60 schools (54 public and six private) were selected across eight states. Up to 52 students were sampled in each school. The anticipated number of assessed students is 1,100 (one thousand one hundred) for the 2029 field trial.

Deliverables:

- 5.1.1 Draft U.S. field test sampling plan
- 5.1.2 Final U.S. field test sampling plan
- 5.1.3 U.S. field test sampling frame, forms, and required data and documentation for submission to the International Sampling Contractor (Core C, Table 3)

5.2 Sampling for the main study

Under this task, the contractor shall prepare a main study sampling plan and fulfill school- and student-level sampling activities described below for the main study including the preparation of U.S. main study sampling frame, forms, and required data and documentation for submission to the International Sampling Contractor (Core C, Table 3). The sampling plan should describe the methodology for drawing the complex stratified sample for the main study.

For school-level sampling, the contractor shall

- work closely with the International Sampling Contractor (Core C, Table 3) to ensure that the school sample design meets all requirements
- provide the school sampling frame (a comprehensive file of schools that includes the entire PISA student population, built from NCES' Common Core of Data (CCD) and Private School Universe Survey (PSS)) to the International Core C sampling contractor.

For student-level sampling, the contractor shall

- participate in the sampling training sessions
- obtain the lists of PISA-eligible students from each participating school and conducting quality control and plausibility checks
- use the within-school sampling software provided by the International Contractor to select the student sample in each school
- monitor the school and student participation
- acknowledge receipt of sampling queries from the International Contractor within one working day and respond to such queries within 5 working days.

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In short, the contractor will provide the school sampling frame to the International Core C sampling contractor, the Core C sampling contractor will sample the schools to participate in the study and provide the list of sampled schools the NC and the NC contract will then select the student sample in each sampled school.

Note that PISA 2029 will introduce a balanced design to provide more robust results and better trend measures across the three core domains of reading, mathematics and science. The design was decided as part of the change from a 3- to a 4-year cycle for PISA. The balanced design involves an equal allocation of testing time among the three core domains. The balanced design requires an increase of the sample by 20 schools per country/economy. This means the minimum number of participating schools increases from 150 in previous cycles to 170 in PISA 2029.

A non-response bias analysis might be required if the achieved response rates for original sampled schools or respondents does not meet the PISA Technical Standards and NCES Statistical Standards. Under this task monitor the contractor shall conduct a non-response bias analysis, if required, based on the proposed approach and methodology in its accepted offer in accordance with these standards.

Deliverables:

- 5.2.1 Final U.S. main study sampling plan
- 5.2.2 Final U.S. main study sampling plan
- 5.2.3 U.S. main study sampling frame, forms, and required data and documentation for submission to the International Sampling Contractor (Core C, Table 3)
- 5.2.4 Draft non-response bias analysis plan
- 5.2.5 Final non-response bias analysis plan
- 5.2.6 Draft non-response bias analysis report
- 5.2.7 Final non-response bias analysis report

Task 6. Data collection

Under this task, the contractor shall recruit and work with School Coordinators and a team of Test Administrators to implement data collection for both the field trial and the main study.

The School Coordinator (SC) is a school staff member (who might be a deputy principal, teacher or administrator) who handles all communication to coordinate the school's participation in PISA. The SC should have access to complete school student records and the authority to schedule testing sessions. The work of SCs is guided by the School Coordinator's Manual, prepared by the International Contractors with some minor adaptation needed for data collection in the U.S.

See Appendix D for the 2025 PISA Computer Based Assessment Preparation Manual. The following manuals (developed by OECD to be provided to the contractor) will give detailed information that schools, and test administrators will need to implement in the field trial, and eventually the main survey:

- PISA 2029 Field Operations Manual
- PISA 2029 School Coordination Manual
- PISA 2029 School Associate Manual

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- PISA 2029 Test Administration Manual
- PISA 2029 Test-taker Delivery System Manual

Note that NCES is planning to collect data using school devices for the PISA 2029 cycle, starting with the field trial. Given the continued challenges in participation rates in 2025, and the additional demand on school staff due to use of school devices NCES is considering increasing school and student incentives as follows for 2029:

- each school = up to \$1500 (\$250 in 2025 cycle)
- each school coordinator = Up to \$500 (\$200 in 2025 cycle)
- each student = Up to \$80 (\$25 in 2025 cycle) and 4 hours of volunteer time.

6.1 Field trial data collection

Under this task, the contractor shall recruit and work with School Coordinators (one per school), recruit and train a sufficient number of Test Administrators proposed in its accepted offer and implement data collection for the field trial that will take place in Spring of 2028 in the U.S. using school devices (online or offline modes). The contractor shall conduct school readiness checks using the School Readiness Tool that will be provided by the OECD (see Appendix E for PISA 2025 School Readiness Tool User Guide).

As a reference, as noted in 4.1 above, for the 2025 field trial a sample of 60 schools (54 public and six private) were selected across eight states. Up to 52 students were assessed in each school. As noted in 5.1, the field trial sample will be a convenience sample.

Deliverables:

- 6.1.1 Draft recruitment and training materials for the field trial
- 6.1.2 Final recruitment and training materials for the field trial
- 6.1.3 Draft data collection plan for the field trial
- 6.1.4 Final data collection plan for the field trial
- 6.1.5 Summary of school readiness checks based on School Readiness Tool
- 6.1.6 Summary of data collection activities for the field trial

6.2 Main Study data collection

Under this task, the contractor shall recruit and work with School Coordinators (one per school), recruit and train a sufficient number of Test Administrators proposed in its accepted offer and implement data collection for the main study that will take place in Spring of 2029 in the U.S. using school devices (online or offline modes). As a reference, 300 schools were sampled in 2025 to exceed the minimum requirement of 150 schools. As noted in 5.2 above, the minimum requirement is 170 schools for 2029 assessments. Up to 52 students will be assessed in each school.

Deliverables:

- 6.2.1 Draft recruitment and training materials for the main study
- 6.2.2 Final recruitment and training materials for the main study

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- 6.2.3 Draft data collection plan for the field trial
- 6.2.4 Final data collection plan for the field trial
- 6.2.5 Summary of school readiness checks based on School Readiness Tool
- 6.2.6 Summary of data collection activities for the main study

6.3 Main Study data processing and disclosure risk review

Before the main study data can be delivered to the OECD, NCES requires that the complete set of data files undergo a disclosure risk review (see appendix C for more details). The contractor shall be aware of and implement the disclosure risk review requirements established by NCES and IES, and prepare a disclosure review plan for review by the IES DRB. The purpose of the disclosure risk review is to identify cases that, due to unique responses, present a high likelihood of respondent identification without some form of data perturbation. The COR or NPM will provide the contractor with an example disclosure risk review document to guide the contractor's efforts. Once the IES DRB has approved the proposed disclosure risk review plan, the contractor shall implement the data perturbations (e.g., swapping, masking) required by the DRB. Only the DRB-approved version of the main study data file may be transferred to the OECD.

In addition, the contractor will be responsible for preparing and completing any OECD information requests about the study, data collection, data files, U.S. operations and procedures, or the U.S. education system, including for states.

- 6.3.1 Draft Disclosure Avoidance Plan for the main study
- 6.3.2 Final Disclosure Avoidance Plan for the main study
- 6.3.3 Draft Disclosure Avoidance Report for the main study
- 6.3.4 Final Disclosure Avoidance Report for the main study

Task 7. Human scoring of cognitive items and coding of responses to questionnaire items

Under this task, the contractor shall recruit and train scorers and score open-ended (constructed-response) cognitive items that cannot be computer-scored in accordance with the PISA Technical Standards. For each of the domains a Lead Scorer is needed. The Lead Scorer receives training from the International Contractors and are responsible for managing and quality assurance of coding in accordance with this and the domain-specific Coding Guide, which the International Contractors develop.

The contractor shall recruit and hire staff to complete scoring no later than 1 month before scoring begins and establish and document rigorous procedures to achieve high scoring reliability in accordance with the PISA Technical Standards. The contractor shall give special consideration to recruiting staff with prior experience in similar efforts, particularly in comparable international assessments or NAEP mathematics, science, and reading assessments. The contractor should also make an effort to recruit and work with the same scorers in both the Field Trial and Main Survey to benefit from their Field Trial experience.

7.1 Human scoring of the field test cognitive items

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Under this task, the contractor shall recruit and train a sufficient number of scorers proposed in its accepted offer and score open-ended (constructed-response) field trial items that cannot be computer-scored in accordance with the PISA Technical Standards. The scoring training is tentatively scheduled to take place **in person** in January 2028. Table 6 shows the anticipated number of items to be hand scored in the field trial.

Table 6. Anticipated number of open-ended field trial items to be scored

Domain	Anticipated number of items
Mathematics	35 [all trend items]
Reading	90 [45-50 items will be trend items]
Science	55 [all trend items]
Media and Artificial Intelligence Literacy	10 [all new items]
Financial Literacy	24 [16 items will be trend items]

The contractor shall deliver the scored data to the international contractor (Core A) in accordance with the timeline specified by OECD or the international contractor. As a reference, twelve coders were recruited for the PISA 2025 field trial.

Deliverables:

7.1.1 Final scored data and documentation for the field trial to be submitted to the International Sampling Contractor (Core A, Table 3)

7.2 Scoring of the main study cognitive items

Under this task, the contractor shall recruit and train a sufficient number of scorers proposed in its accepted offer and score open-ended (constructed-response) main study items that cannot be computer-scored in accordance with the PISA Technical Standards. The scoring training is tentatively scheduled to take place **virtually** in January 2029. Table 7 shows the anticipated number of items to be human scored.

Table 7. Anticipated number of open-ended main study items to be scored

Domain	Anticipated number of items
Mathematics	35
Reading	77
Science	55
Media and Artificial Intelligence Literacy	5
Financial Literacy	16

The contractor shall deliver the scored data to the international contractor (Core A) in accordance with the timeline specified by OECD or the international contractor. As a reference, thirty-four coders were recruited for the PISA 2025 main study.

Deliverables:

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7.2.1 Final scored data and documentation for the main study to be submitted to the International Sampling Contractor (Core A, Table 3)

7.3 Coding responses to fill-in item in questionnaires

Under this task, the contractor shall code students' responses to the fill-in questions about their parents' occupation (for mother, father, parent/guardian #1 and parent/guardian #2) based on the International Standard Classification of Occupations (ISCO)-08 classification system. As a reference, responses to these questions both the field trial and the main study U.S. data were human coded by three coders in 2025 cycle, with 25% of the responses double coded for reliability checks.

Deliverables:

7.3.1 Final coded student questionnaire data for the field trial to be submitted to the International Sampling Contractor

7.3.2 Final coded student questionnaire data for the main study to be submitted to the International Sampling Contractor

Task 8. Teacher Questionnaires (Optional)

Under this task, the contractor shall conduct all sampling and data collection activities for the Teacher Questionnaire based on the proposed approach and methodology in its accepted offer in accordance with the PISA Technical Standards. More specifically, the contract shall

- complete sampling forms related to the teacher target population
- collect information on eligible teachers in the schools (teacher lists), and upload the information to the sampling platform provided by the international contractor or OECD
- prepare materials to inform teachers about the questionnaire to encourage maximum participation
- provide teachers with online links along with their access codes
- monitor participation rate and implement survey operation procedures
- conduct data cleaning and processing post administration, including steps necessary to avoid disclosure risk

The Teacher Questionnaire will be administered to all eligible teachers in the schools that are sampled to participate in PISA (no within-school sampling of teachers). The questionnaire will be highly focused and shorter than in previous cycles (around 15-20 minutes) to facilitate higher response rates. The questionnaire will be computer-based and administered online.

Before the Teacher Questionnaire data can be delivered to the OECD, NCES requires that the complete set of data files undergo a disclosure risk review (see NCES Statistical Standards available at <https://nces.ed.gov/statprog/2012/>). The contractor shall be aware of and implement the disclosure risk review requirements established by NCES and IES and prepare a disclosure avoidance plan for review by the IES Disclosure Review Board. The COR or NPM will provide the contractor with an example Disclosure Avoidance Plan (DAP) to guide the contractor's

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efforts. Once the IES DRB has approved the proposed disclosure risk review plan, the contractor shall implement the data perturbations (e.g., swapping, masking) required by the DRB. The contractor shall also prepare a report summarizing the results of the data perturbations in a Disclosure Avoidance Report. Only the DRB-approved version of the main study data file may be transferred to the OECD.

Deliverables:

- 8.1 U.S. Teacher Questionnaire sampling frame, forms, and required data and documentation for submission to the International Sampling Contractor (Core C, Table 3)
- 8.2 Draft recruitment materials for the U.S. Teacher Questionnaire data collection
- 8.3 Final recruitment materials for the U.S. Teacher Questionnaire data collection
- 8.4 Draft Disclosure Avoidance Plan
- 8.5 Final Disclosure Avoidance Plan
- 8.6 Draft Disclosure Avoidance Report
- 8.7 Final Disclosure Avoidance Report
- 8.8 Summary of U.S. Teacher Questionnaire data collection activities

Task 9. Main study electronic data files and user's guides

For purposes of disseminating data for the study, the contractor shall prepare and deliver an electronic package with the main study U.S. data containing:

- The U.S. dataset in restricted-use form, with internationally restricted variables and additional U.S.-specific variables included (such as race/ethnicity, or other variables not included in the international dataset), in ASCII, so that it can be read into any statistical analysis software.
- READ.ME files that provide guidelines for accessing the data files and that describe the data files included in the released package.
- Variable codebooks that provide a comprehensive list of every variable in the data file(s).
- Record layout files to assist data users with quickly locating the variable(s) of interest in the data files.

The contractor shall produce a U.S.-version of the restricted-use datasets for release through NCES. The construction of these dataset files will be modeled after those files previously released for the study. The datasets produced under this task must make use of the dataset approved by the NCES Data Review Board (DRB).

Deliverables:

- 9.1 Draft restricted-use main study data file package
- 9.2 Final restricted-use main study data file package

Task10. Teacher Questionnaire electronic data files and user's guides (Optional)

For purposes of disseminating data for the study, the contractor shall prepare and deliver an electronic package with the Teacher Questionnaire data containing U.S. dataset in restricted-use

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form along with READ.ME files, a codebook and record layout file, all similar to those listed under Task 9.

Deliverables:

- 10.1 Draft restricted-use Teacher Questionnaire data file package
- 10.2 Final restricted-use Teacher Questionnaire data file package

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Appendix A. PISA 2029 Tentative Schedule and Background Information for National Preparations

1. This document aims to support national preparations and resource allocation for PISA 2029. The information in the document is tentative and subject to change as the international preparations continue during 2025 and 2026.
2. The document builds on the PISA Governing Board's decisions regarding PISA 2029, in particular the change to a balanced design with the change to a 4-year cycle, as agreed at the 54th PGB meeting, the strategy for international options in PISA that sets out which options to offer in this cycle, as agreed at the 47th PGB meeting, and the PISA long-term strategy for 2024-33.

Assessment domains

3. PISA 2029 will continue to focus on three core domains: reading, mathematics and science. Reading will be the “focal domain”, which is similar to the previous “major domain”, except that it will not be given a larger share of the sample allocation than the other two domains, as PISA is moving to a balanced design. The framework and test items will be updated for reading, the contextual questionnaires will focus on questions related to the teaching and learning of reading, and the international reports will focus on analysing the relationships between contextual factors and student performance in reading.
4. The innovative domain for PISA 2029 is Media and Artificial Intelligence Literacy (MAIL). It is currently defined as the set of competencies to interact with digital content and platforms effectively, ethically and responsibly. The goals of the assessment are to strengthen our understanding of the knowledge, skills and attitudes that students need to effectively and responsibly consume and produce digital content, and to obtain internationally comparable data on how students use digital content and AI platforms, and whether they reflect on their actions before making decisions in digital spaces.

Survey and sample design

5. PISA 2029 will introduce a balanced design to provide more robust results and better trend measures across the three core domains of reading, mathematics and science. The design was decided as part of the change from a 3- to a 4-year cycle for PISA. The balanced design involves an equal allocation of testing time among the three core domains.
6. For national centres, the balanced design requires an increase of the sample by 20 schools per country/economy. This means the minimum number of participating schools increases from 150 in previous cycles to 170 in PISA 2029. The minimum sample size for adjudicated regions will remain unchanged.

International options

7. PISA 2029 will offer four international options:
 - Financial Literacy
 - Emerging Policy Questionnaire

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- Teacher Questionnaire
- Parent Questionnaire

8. Countries/economies will sign up for the international options during 2026. Countries/economies will be asked to formally commit to their participation in international options by August 2026. International options are offered subject to sufficient demand, usually a minimum of 10 countries/economies per option.

Contractors

9. As in previous cycles, the development and international co-ordination of PISA 2029 will be carried out by a group of contractors. The contractors are selected by the OECD following a competitive procurement process, based on the terms of reference and selection criteria determined by the PISA Governing Board.

10. The contracts are currently under preparation.

11. The OECD has selected the following organisations as the preferred contractor for the three core components:

Core A: Main contractor for the development, implementation and co-ordination

- ACER (Australian Council for Educational Research) in a consortium with cApStAn and HallStat

Core B: Innovative domain development

- ACER (Australian Council for Educational Research)

Core C: Sampling

- IEA

Schedule of tasks for National Centres

12. Table 1 provides the tentative schedule of activities for the implementation of PISA 2029 in participating countries/economies. The schedule is adapted to the international testing window for PISA 2029 determined by the PISA Governing Board at its 58th meeting in November 2024 based on a consultation among all participating countries and economies. The schedule and list of activities are subject to change and will be further developed as the contractors commence their work in 2026.

Table 2. Schedule of tasks for National Centres in PISA 2029

All dates and activities are tentative. The final schedule will be confirmed in 2026 as part of the NPM manual.

Phase	Period (approximate)	Tasks for PISA National Centers
Development phase	Jan-26 to Dec-26	Establish PISA National Centre and appoint National Project Manager Participate in item development workshop (optional) Contribute ideas for new test items (optional) Review new test items
Field Trial preparation	Jan-27 to Jan-28	Translate and adapt test instruments and manuals Review and test national version of Field Trial instruments Sampling and recruitment of schools for the Field Trial Recruitment and training of test administrators Coder training
Field Trial data collection <i>Data collection period: 1 February to 15 September 2028</i>	Feb-28 to Sep-28	Support schools to prepare and plan for testing Co-ordinate activities of test administrators Distribute materials to schools Organize and monitor data collection Coding of responses Data management and data submission
Field Trial review phase	Aug-28 to Oct-28	Review Field Trial data analysis reports Review Field Trial materials and request changes to errors Review operational procedures
Main Survey material preparation	Jun-28 to Sep-28	Main Survey - Translate and adapt test instruments
Main Survey operational preparation	Oct-28 to Jan-29	Preparation of manuals and school-level materials Review and test national version of Main Survey instruments Sampling and recruitment of schools for the Main Survey Recruitment and training of test administrators Coder training
Main Survey data collection <i>Data collection period: 1 February to 14 September 2029</i>	Feb-29 to Sep-29	Support schools to prepare and plan for testing Co-ordinate activities of test administrators Distribute materials to schools Organize and monitor data collection Coding of responses Data management and data submission
Report preparation phase	Sep-29 to Aug-30	Review national databases Review international databases Analyze national data and prepare national report Prepare dissemination of results

International meetings and trainings

13. The PISA 2029 contractors will organise a series of meetings and trainings for National Project Managers and relevant team members. Participation in the meetings and trainings is required and essential for all countries/economies to conduct PISA successfully and adhere to the technical standards. Participation is required both for new and experienced National Centres as each cycle involves important updates to the test platform, procedures and instruments. Countries/economies are invited to allocate funding for in-person participation in the meetings and training.

Table 3. Schedule of meetings for National Centres in PISA 2029

Timeline (approximate)	Meeting and format
March 2027	<i>NPM Meeting 1 (in person)</i>
September/October 2027	<i>NPM Meeting 2 (in person)</i>
January 2028	<i>Field Trial coder training (in person)</i>
September/October 2028	<i>NPM Meeting 3 (in person)</i>
January 2029	<i>Main Survey coder training (virtual)</i>

Schedule of key documents

14. Participating countries/economies will receive the first technical documentation for PISA 2029 during 2026 as the international contractors commence the preparations for this cycle. Table 3 shows the tentative schedule of key documents during the preparation phase. These documents can help Ministries and Agencies in charge of PISA 2029 to establish their National Centres, determine the resources needed for the implementation of the survey nationally, and support national procurement processes where needed.

Table 4. Schedule of key documents for National Centres in PISA 2029

Timeline (approximate)	Document
May 2026	<i>Requirements for participation in PISA 2029</i> This document will provide a technical overview of PISA 2029, including a description of the equipment needed for PISA 2029 and the devices that will be supported. The document will also include a detailed overview of the resources needed to establish the National Centre (NC) for PISA 2029 and administer the test. It will describe the roles and responsibilities of the National Project Manager (NPM) and NC, including suggestions for specific staff roles within the NC. <i>Working with the ACER consortium</i> This document will outline the Consortium's organisational structure and their roles, and it will specify the consortium's intended working relationships with NPMs.
Oct 2026	<i>NPM Manual</i> This first edition of the NPM Manual will provide detailed instructions for the tasks needed from the start of the PISA 2029 cycle through to the Field Trial. It will include the schedule of tasks for National Centres for this period.
November 2026	<i>Technical Standards</i> The Technical Standards are updated for each PISA cycle and agreed by the PISA Governing Board. The contractors will review the existing standards and prepare a draft document with the relevant updates by March 2026 for an initial discussion at the PGB meeting in April 2026. The final draft of the Technical Standards will be presented at the PGB meeting in October/November 2026 for review and approval.

Appendix B. Data Confidentiality, Data Security Plan, and Affidavit of Nondisclosure, and Contractor Employee Security Screening Requirements

Confidentiality of Individuals and Institutions

NCES assures participating individuals and institutions that any data collected conforms to the IES standards for protecting the privacy of individuals as required by 20 U.S.C., § 9573):

“... all collection, maintenance, use, and wide dissemination of data by the Institute, including each office, board, committee, and Center of the Institute, must conform with the requirements of section 552A of Title 5, United States Code [which protects the confidentiality rights of individual respondents with regard to the data collected, reported, and published under this title].” (20 U.S.C., § 9573)

Under the Education Sciences Reform Act of 2002 (ESRA 2002), all individually identifiable information about students, their families, and their schools must remain confidential. To this end, this law requires that no person may:

1. Use any individually identifiable information furnished under the provisions of this section for any purpose other than statistical purposes for which it is supplied, except in the case of terrorism;
2. Make any publication whereby the data furnished by any particular person under this section can be identified; or
3. Permit anyone other than the individuals authorized by the Commissioner to examine individual reports.

Further, individually identifiable information is immune from legal process, and must not, without the consent of the individual concerned, be admitted as evidence or used for any purpose in any action, suit, or other judicial or administrative proceeding, except in the case of terrorism. Employees, including temporary employees, or other persons who have sworn to observe the limitations imposed by this law, who knowingly publish or communicate any individually identifiable information will be subject to fines of up to \$250,000 or up to five (5) years in prison, or both (Class E felony).

Information Collection Request (ICR)

All respondents asked to participate in data collection initiatives must be informed of the following:

- 1) NCES enabling legislation which authorizes the information collection,
- 2) purposes for which the information is needed,
- 3) uses that may be made of the information,
- 4) that data will not be reported in a way to reveal individual responses,
- 5) participation is voluntary, and
- 6) no adverse action will result if the requested information is not provided.

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Survey respondents, if they so request, must be given access to their records and be permitted to amend their responses.

Data Security and Data Security Plan

Definition of Personally Identifiable Information and Direct Identifiers

The term “personally identifiable information” (hereinafter PII) means any information about an individual maintained by an agency, including, but not limited to, education, financial transactions, medical history, and criminal or employment history and information which can be used to distinguish or trace an individual’s identity, such as their name, social security number, date and place of birth, mother’s maiden name, biometric records, etc., including any other personal information which is linked or linkable to an individual. The term “direct identifier” denotes any single datum that alone is deemed sufficient to yield a high risk of disclosure if released, such as social security numbers, full names, or biometric records.

Development of Data Security Plan

The contractor must present a detailed security plan that expands upon what was presented in the proposal to the Task Leader/COR for approval. In order to ensure the anonymity of individual respondents, the contractor must draft and execute this plan in compliance with all relevant laws, regulations, and policies governing the security of data, particularly PII. These include (but are not limited to):

1. 20 U.S.C., § 9573 of the Education Sciences Reform Act of 2002, which states:
 - (a) IN GENERAL - All collection, maintenance, use, and wide dissemination of data by the Institute, including each office, board, committee, and center of the Institute, must conform with the requirements of section 552a of title 5, United States Code, the confidentiality standards of subsection (c) of this section, and sections 444 and 445 of the General Education Provisions Act (20 U.S.C. 1232g, 1232h).
 - (b) STUDENT INFORMATION - The Director must ensure that all individually identifiable information about students, their academic achievements, their families, and information with respect to individual schools, must remain confidential in accordance with section 552a of title 5, United States Code, the confidentiality standards of subsection (c) of this section, and sections 444 and 445 of the General Education Provisions Act (20 U.S.C. 1232g, 1232h);
2. The Privacy Act of 1974 (5 U.S.C. 552a);
3. Cybersecurity Enhancement Act of 2015 (6 U.S.C. §151);
4. Foundations of Evidence-Based Policymaking Act of 2018, Title III, Part B, Confidential Information Protection;
5. The U. S. Department of Education OCIO-1 Handbook for Information Assurance Security Policy (January 2009);

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6. The U.S. Department of Education Incident Handling Procedures (February 2009); and
7. Office of Management and Budget Memorandum dated 1/03/2017 on Preparing for and Responding to a Breach of Personally Identifiable Information (M-17-12).

The data security plan will, at a minimum, cover the following issues: the transfer/transport of PII, maintenance, storage, and use of direct identifiers, replacement of direct identifiers with internal codes, security of master survey files, security and confidentiality training for contractor staff, and reporting of data security breaches. Hence, the data security plan must be written to place particular emphasis on the following aspects of confidentiality and security:

Guidelines for Transfer of PII

As a general principle, the physical and/or electronic transfer of PII (particularly direct identifiers) must be limited to the extent practicable. This limitation includes both internal transfers (e.g., transfer of information between agents of the contractor, including subcontractors and/or field workers) and external transfers (e.g., transfers between the contractor and the agency, or between the contractor and another government agency or private entity assisting in data collection).

When it is necessary for the contractor to transfer PII, such transfer must occur in a manner that maximizes the security of information. The data security plan must describe the likely modes of data transfer and establish the most secure procedures practicable. The most secure method of transferring sensitive information is not necessarily the most efficient or convenient one.

1. The preferred means of transferring PII (and particularly direct identifiers) is electronic transfer with 128-bit secure sockets layer (SSL) encryption.
2. If electronic PII is transferred via physical media (e.g., a CD-ROM or DVD-ROM is created and sent via courier service or mail), all files on the media must be encrypted (FIPS 140-2 standard) with a strong password (e.g., a random string of letters and numbers) and this password must be sent separately from the data. In addition, no obvious external markings or information identifying the media as containing PII must be written, stamped, or otherwise inscribed on the media or its packing material.
3. If paper records or files must be transferred, then the contractor and its agents should hand carry these documents for delivery. Shipping or mailing these kinds of documents should be avoided. The contractor should clearly state in their data security plan if this requirement cannot be met. The data security plan should describe all transfers of paper records or files and how they will be secured. If these documents must be shipped/mailed, the contractor and its agents should avoid transferring multiple direct identifiers in the same shipment. For example, a mailing containing the names, addresses, and social security numbers of subjects is less secure than two mailings: one shipment with the names and addresses and a second shipment with the social security numbers and no additional information. Alternatively, a partial mailing of data could be followed up with a telephone call

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(for smaller amounts of PII) which provides the remainder of the data verbally. Any shipment sent must have a tracking number. As above, no obvious external markings or information identifying the mailing as containing PII must be written, stamped, or otherwise inscribed on the packing material.

Maintenance, Storage, and Use of Direct Identifiers

Under no circumstances may the contractor release PII to unauthorized individuals. In addition, direct identifiers must be maintained in files which are physically separate from other research data, and which are accessible only to sworn agency and contractor personnel. Individual direct identifiers used during the course of the project must be associated with data only for statistical purposes such as data gathering, matching new data with old, establishing sample composition, authenticating data collections, editing data based on callbacks, or obtaining missing information.

As a general principal, the number of personnel with access to PII (and particularly direct identifiers) must be restricted to the smallest number as possible.

Replacement of Direct Identifiers with Internal Codes

Although direct identifying information is often required in the data collection process, to the extent practicable the contractor must use internally generated identifying codes to track subjects in data files. Whenever possible, the transfer and storage of files containing multiple direct identifiers should be avoided. If the transfer of direct identifiers is required for statistical purposes such as data matching, the minimum number of direct identifiers needed for successful matching must be used.

Security of Master Survey Files

The contractor must maintain security on the complete set (and deliverable backups) of all master survey files and documentation.

IT Security and Policies

The contractor must:

- Complete/update the appropriate level of Security Accreditation (SA) documentation per NIST Risk Management Framework guidance, security controls testing, interagency security agreements (ISAs), and risk assessments in support of government issuance of security assessment and authorization to operate (ATO) decisions
- Ensure that systems/products/applications have the ability to facilitate single-sign-on capabilities and required support for HSPD 12 Personal Identity Verification (PIV) enablement and integration
- Include the capability for network traffic that flows between externally hosted systems and networks, to/from Department systems and networks, to be routed through one of the Department's Trusted Internet Connections (TIC) gateways as part of the solution configuration. Implement controls to ensuring all possible traffic, including mobile and

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cloud, goes through a TIC. Implement connections between Department systems and networks with externally hosted systems that are in compliance with the requirements of the Trusted Internet Connections (TIC) initiative

- Architect contractor hosting environments to use security isolation and network segmentation principles in order to ensure that the environments are properly protected against an unauthorized access and threat from adversaries who may strive to move laterally across internal Department or contractor hosted systems and network segments.
- Provide an automated capability and process to scan and assess all systems and assets, and associated logs for malicious indicators of compromise (IOCs) identified by the Department regarding priority threat-actor Techniques, Tactics, and Procedures (TTPs); the contractor is required to have the capability to scan for indicators of compromise within 24 hours of receipt of the indicators provided by the Department of Education from the Department of Homeland Security
- Implement and maintain capabilities and processes to support the timely detection of, reporting, and rapid response and recovery to cyber incidents in accordance with timelines and requirements specified in Federal guidance and Department cybersecurity incident reporting policy guidance
- In support of cybersecurity performance measure reporting, the contractor must implement and maintain an automated software asset management/inventory and hardware asset inventory capability (e.g., scans/device discovery processes) at the enterprise-level.
- Implement capabilities to rapidly deploy emergency security patches and implement specific security control enhancements as directed by the Department of Homeland Security to all Federal Departments via mechanisms such as the DHS Cybersecurity Coordination, Assessment, and Response (C-CAR) action items, and DHS Binding Operational Directives (BODs).
- Implement capabilities and processes to patch all critical vulnerabilities identified to the Department of Education by DHS immediately or, at a minimum, within 30 days of patch release.
- Ensure robust physical and cybersecurity protections are in place for all of the Department's high value assets (HVAs). The identification of HVAs by the Department will be an ongoing activity due to the dynamic nature of cybersecurity risks.
- Implement remote access solutions that only use multi-factor authentication solutions and that prohibit the use of split tunneling and/or dual-connected remote hosts where the connecting device has two active connections.
- Implement remote access solutions that scan for malware before allowing full connections and that time out after 30 minutes (or less) of inactivity and require re-authentication to re-establish a session
- Implement capabilities for all incoming email traffic to pass through anti-phishing and anti-spam filtration at the outermost border mail agent or server
- Implement capabilities for all incoming email traffic to be analyzed using sender authentication protocols (e.g., DKIM, DMARC, VBR, SPF, iprev)
- Implement capabilities that ensure that incoming email traffic is analyzed using a reputation filter (to perform threat assessment of sender)
- Implement capabilities that ensure that incoming email traffic is analyzed for detection of clickable URLs, embedded content, and attachments; and incoming email traffic is first

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analyzed for suspicious or potentially nefarious attachments and opened in a sandboxed environment or detonation chamber

- Implement capabilities for all outbound communications traffic to be checked at the external boundaries to detect encrypted exfiltration of information (i.e., capability to decrypt/interrogate and re-encrypt)
- Implement effective network segmentation design and security solutions to limit potential threats from adversaries attempting lateral movement across systems on the Department's (or contractor's networks), and also to better protect and securely isolate the Department's HVAs
- Implement and maintain Information Security Continuous Monitoring (ISCM) and Continuous Diagnostics and Mitigation (CDM) capabilities for all IT assets to be subject to an automated inventory, configuration, and vulnerability management capability, with real time reporting
- Implement and maintain strong authentication capabilities requiring the technical enforcement of all users being required to use a Personal Identity Verification (PIV) card to authenticate to the network, (with exceptions for a very limited set of users specifically approved by the Department)
- Develop and maintain (or update existing) System Security Plans (SSP) and security controls assessment (SCA) test plans for the network general support system (GSS), and infrastructure systems
- Provide support to creating the security assessment and authorization (or accreditation) (SA&A) packages and documentation in accordance with the Risk Management Framework guidance and processes specified by NIST and Department guidance
- Implement security configurations on all IT assets and systems using DISA STIGs and other industry recognized best practices or guidance
- Perform security configuration management to include configuring all Windows based systems with the latest United States Government Configuration Baseline (USGCB) security settings available from the NIST website
- Support annual or emergent security audits and security scans that may be performed by the Office of Inspector General (OIG), the General Accountability Office (GAO), or the Department of Homeland Security (DHS)
- The contractor must provide availability and accessibility to the Department, to the OIG, and to any third party vendors designated by the Department to: 1) Review audit findings; 2) Determine if corrective actions were properly implemented and the associated audit findings were properly closed; 3) Support cybersecurity incident analysis and forensics activities.
- Produce scheduled Monthly/Quarterly/Annual security performance measure reports that align to the Department's cybersecurity performance measure reporting requirements specified by OMB for FISMA, the President's cybersecurity Cross Agency Priority (CAP) goals and targets, and CyberScope reporting. Security performance measure reports must use the format and template specified in the Annual CIO FISMA metrics specified by OMB and DHS.
- Provide for the encryption for PII, CUI, Data at Rest and data in transit, Encryption solutions applied must be FIPS 140-2 validated.
- Document and track contractor personnel cyber training based on roles

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- Develop, maintain, and publish a listing of Contractor-provided security controls, hybrid security controls, and “customer”-provided security controls, in support of systems security assessments and authorizations, and the issuance of authority to operate (ATO) decisions by the Department.
- Provide security audit support (e.g., A-123), including scheduled and event-driven audits.
- Capture and provide forensic disk images to support security incident analysis, malware analysis, or other investigative requirements (such as specific requests from the OIG or law enforcement)
- Provide support for threat monitoring and analysis, incident response, vulnerability management, risk management, continuous monitoring and reporting and other traditional security operations center activities
- Provide and maintain multi-factor authentication solutions utilizing the Personal Identity Verification (PIV) card (or a Department- approved Level of Authentication -4 solution); and utilize FIPS 140-2 approved encryption for all remote access requirements
- Provide robust encryption capabilities to include services such as digitally signed and encrypted email, and default encryption for sensitive information held by the Department. Solutions should be available to enable encryption of as much data at rest and data in transit as possible.
- Identify, perform, track, and report vulnerability and security weakness remediation and mitigation activities through the Department’s Plan of Action and Milestones process (POA&M) in accordance with Departmental information security policy
- Establish, maintain, and execute standard configuration management processes for all cybersecurity software and hardware
- Implement and maintain a Privileged Account Management Solution to improve the identity and access management of user accounts, while also meeting Department targets to tightly control and limit the number of users with elevated privileges
- Implement and maintain tightened processes for managing privileged user accounts, to include implementation of capabilities to limit functions that can be performed when using privileged accounts; limit the duration that privileged users can be logged in; limit the privileged functions that can be performed using remote access; prohibit Internet access when privileged users are performing systems administrations tasks; and ensure that privileged user activities are logged and regularly reviewed
- Document and maintain system security boundaries, system configuration details, and network diagrams, in support of security assessment and authorization to operate (ATO) processes
- Develop and implement processes for revising system security documentation on a scheduled and event-driven basis
- Provide support for maintaining system security documentation in support of FISMA reporting requirements and security compliance status in the Department’s Cyber Security Assessment and Management (CSAM) system
- Develop and submit system security documentation, risk assessments, security controls testing reports, and any required privacy impact analysis (PIA) to the Department in support of the Risk Management Framework processes and ATO decisions for IT environment components

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- Develop corrective/remediation plans of action and milestones (POAMs) and strategies to address security audit and assessment findings, and other reports of system security weaknesses or non-compliance
- Develop and maintain a system security architecture; the contractor's solution must include effective network segmentation design and solutions to limit lateral movement across systems on the Department's networks, and also better protect the Department's HVAs
- Utilize PIV or other approved Level of Assurance 4, as defined in NIST SP 800-63-2 Electronic Authentication Guidelines, compliant Identity and Access Control mechanisms for network/domain administrative enterprise access.
- Maintain near real-time security monitoring and intrusion detection capabilities to enable the contractor and the Department to know the security risk posture of the network at any given time;
- Configure all Windows based systems with the latest United States Government Configuration Baseline (USGCB) security settings available from the NIST website
- Utilize multi-factor authentication, including integration and compliance with HSPD-12 PIV requirements, for all remote access solutions for the Department's sensitive information systems
- Provide a multi-tier disaster recovery capability that provides the infrastructure and process to meet the recovery requirements of all of its High Value Assets (HVAs), and applications (Mission-Critical, Decision Support, Other)
- Provide IT Disaster Recovery Planning and Management capabilities and support
 - Define business risk and risk assessment
 - Develop disaster recovery strategies
 - Develop disaster recovery plans
 - Develop IT system contingency plans
 - Conduct disaster recovery exercises, training and awareness
- Provide Disaster Recovery Operational Services, including Contractor support to the Department in the planning, preparation, implementation, and documentation of a Disaster Recovery Program that includes the capabilities described below:

The contractor, and all sub-contractors, must comply with the Department of Education's IT security policy requirements, and other applicable procedures and guidance. The contractor, and all sub-contractors, must develop and implement management, operational and technical security controls to assure required levels of protection for information systems. The contractor, and all sub-contractors, must further comply with all applicable Federal IT security requirements including, but not limited to, the Federal Information Security Modernization Act (FISMA) of 2014, Office of Management and Budget (OMB) Circular A-130, Homeland Security Presidential Directives (HSPD), including HSPD-12, Personal Identity Verification (PIV) Enablement and Integration, and single sign-on, the most recent National Institute of Standards and Technology (NIST) special publications, standards and guidance, and the Federal Risk and Authorization Management Program (FedRAMP) requirements and guidance.

These security requirements include, but are not limited to, the successful Security Assessment and Authorization (SA&A) of the system (includes commercially owned and operated systems managed by the commercial vendor and its sub-contractors, supporting Department programs,

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contracts, and projects); obtaining a full Authority to Operate (ATO) before being granted operational status; performance of annual self-assessments of security controls; annual Contingency Plan testing; performance of periodic vulnerability scans; updating all information system security documentation as changes occur; and other continuous monitoring activities, which may include, mapping, penetration and other intrusive scanning. Full and unfettered access for any of the Department's third party Managed Security Services Provider (MSSP) or Cyber-operations prevention testers, or vulnerability scanners, or auditors must be granted to access all computers and networks used for this system. Additionally, when there is a significant change to the system's security posture, the system (Federal and commercial prime- and sub-contractors included) must have a new SA&A, with all required activities to obtain a new ATO, signed by the Authorizing Official (AO).

System security controls must be designed and implemented consistent with the current, finalized version of the NIST SP 800-53, 'Recommended Security Controls for Federal Information Systems and Organizations.' All NIST SP 800-53 controls must be tested / assessed no less than every 3 years, according to federal and Department policy. The risk impact level of the system will be determined via the completion of the Department's inventory form and must meet the accurate depiction of security categorization as outlined in Federal Information Publishing Standards (FIPS) 199, 'Standards for Security Categorization of Federal Information and Information Systems.'

System security documentation must be developed to record and support the implementation of the security controls for the system. This documentation must be maintained for the life of the system. The contractor, and all sub-contractors, must review and update the system security documentation at least annually and after significant changes to the system, to ensure the relevance and accurate depiction of the implemented system controls and to reflect changes to the system and its environment of operation. Security documentation must be developed in accordance with the NIST 800 series and Department of Education policy and guidance.

The contractor, and all sub-contractors, must allow Department employees (or Department designated third party contractors) access to the hosting facility to conduct SA&A activities to include control reviews in accordance with the current, finalized version of the NIST SP 800-53, and the current, finalized version of the NIST SP 800-53A. The contractor, and all sub-contractors, must be available for interviews and demonstrations of security control compliance to support the SA process and continuous monitoring of system security. In addition, if the system is rated as 'Moderate' or 'High' for FIPS 199 risk impact, vulnerability scanning and penetration testing must be performed on the hosting facility and application as part of the SA&A process. Appropriate access agreements will be reviewed and signed before any scanning or testing occurs.

Identified deficiencies between required security controls within the current, finalized version of the NIST SP 800-53 and the contractor's, and all sub-contractor's implementation, as documented in the Risk Assessment Report, System Security Plan (SSP) and Security Assessment Report (SAR), must be tracked for mitigation through the development of a Plan of Action and Milestones (POA&M) in accordance with Department policy. Depending on the severity of the deficiencies, the Department may require remediation before an ATO is issued.

The contractor must provide cybersecurity strategies, infrastructure hosting environments, and solutions that comply with the requirements of the Federal Information Security Modernization

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Act (FISMA), Department cybersecurity policy guidance, and guidance contained in the NIST Special Publications series such as NIST Special Publication 800-53 and other NIST Special Publications. The contractor must provide solutions that support the Department's efforts to implement and maintain effective protection activities such as reducing the attack surface and complexity of IT infrastructure; minimizing the use of administrative privileges; utilizing strong authentication credentials; safeguarding data at rest and in-transit; training personnel; ensuring repeatable processes and procedures; adopting innovative and modern technology; ensuring strict domain separation of critical/sensitive information and information systems; implementing network segmentation architectures to better protect and isolate the Department's high value assets and most sensitive information and data; and ensuring a current inventory of hardware and software components. The contractor must include actions and initiatives to implement the NIST Cybersecurity Framework that emphasizes and measures capabilities to "Identify, Protect, Detect, Respond, and Recover," and ensure that all applicable Service Level Agreements (SLA)s are adhered to, complied with, and satisfied.

All awarded contracts must ensure that:

1. Their IT product/system is monitored during all hours of operations using entrusted detective/preventive systems;
2. Their IT product/system has current antiviral products installed and operational;
3. Their IT product/system is scanned on a reoccurring basis;
4. Vulnerabilities are remediated in a timely manner on their IT product/system; and
5. Access/view for cyber security situational awareness on their IT product/system is made available to the Department CIRC (cyber incident response capability).
6. All applicable Service Level Agreements (SLA)s are adhered to, complied with, and satisfied.

Information Assurance Requirements

Additional Requirements

As noted above all contractors and subcontractors must comply with the language in the September 2020 "Security and Privacy Requirements for Information Technology Procurements" with exception of, and in addition to:

- E-Government Act of 2002 has been superseded by FISMA 2014 (Federal Information Security Modernization Act) of 2014. All contractors and subcontractors must comply with all FISMA 2014 provisions.
- The Contractor (and/or any subcontractor) must comply with applicable federal laws that include, but are not limited to, the ED Departmental Handbook on Information Assurance / Cybersecurity Policy (OCIO: 3-112), and its subordinate five (5) Policy Framework Instructions / Standards documents, based on the National Institute of Standards and Technology (NIST) "Framework for Improving Critical Infrastructure Cybersecurity."; the FISMA 2014, (44 U.S.C. 101); Office of Management and Budget (OMB) Circular A130, Managing Information as a Strategic Resource; and other applicable federal laws, regulations, Departmental policies, and the most current, finalized and up-to-date NIST Special Publication (SP) 800 series (as amended).
- If there will be any access to Controlled Unclassified Information (CUI), or to the U.S. Department of Education (ED) resources, systems, or equipment, then in accordance with

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The United States Department of Education (ED) Acquisition Alert 2020-01, “Education Acquisition Regulation (EDAR) Class Deviation: Cyber and Personnel Security Requirements for Contractors,” dated December 10, 2019, which is (or will be) posted on the external ED.Gov site, at: <https://www2.ed.gov/policy/fund/reg/clibrary/acquisition-alert-2020-01.pdf> and which is also posted (or will be), on the Department of Education’s Acquisition Alert site, at: <https://connected.ed.gov/ofc/oaga/Pages/Acquisition-Alerts.aspx>. All Contractors and subcontractors must comply with the DEVIATIONs to the two (2) Department Security Requirements clauses, referenced in the following required action: Required Action: Contracting Officers must include the provision, 3452.239-71 “Notice to offerors of Department security requirements (DEVIATION)” in all new solicitations and contracts, and clause, 3452.239-72 “Department security requirements (DEVIATION)” in all new solicitations and contracts, as applicable. Within 90 days contracting officers must modify existing contracts, to include 3452.239-72

- All contractors and subcontractors must comply with requirements that all Department systems, application appliances, or website/web services:
 - Support Elliptic Curve Diffie-Hellman (ECDHE) suites to enable forward secrecy with servers and web browsers. “Forward secrecy”, also known as perfect forward secrecy, is a property of secure communication protocols in which compromises of long-term keys does not compromise past session keys. Forward secrecy protects past sessions against future compromises of private key
 - Reorder system and services cipher suites to ensure the ECDHE suites are prioritized, followed by the Diffie-Hellman suites
 - Configure systems and services to enable other non-Diffie-Hellman key exchange cipher suites from the list of cipher suites offered by the SSL client
- The contractor must comply with the Foreign Investment Risk Review Modernization Act of 2018 (FIRRMA), and with its amendment, the Federal Acquisition Circular (FAC) 2019-05
- Prior to contract award the Contracting Officer (CO) must ensure appropriate compliance with Section 889, Parts A and B of the National Defense Authorization Act (NDAA) Fiscal Year (FY) 2019 for the prohibition of contracting with entities using certain covered Information and Communications Technologies (ICT) and services. Below are the sub-steps required for validation as outlined from [Acquisition Alert 2020-05](#):
 - Review [FAR 52.204-25](#) – Prohibition on Contracting for Certain Telecommunications and Video Surveillance Services or Equipment
 - Complete the representation at [FAR 52.204-24\(d\)\(2\)](#), after conducting a reasonable inquiry, and submit the representation to the CO.
 - If offeror represents “does” in response to the representation at FAR 52.204-24(d)(2), complete the disclosure at FAR 52.204(e)(2) and submit it to the CO.
- Effective March 22, 2021, the Department of Commerce released Title 15 of the Code of Federal Regulations (CFR) Part 7, “Securing the Information and Communications Technology and Services (ICTS) Supply Chain” to implement provisions of Executive Order 13873. Specifically, 15 CFR Part 7 prohibits ICTS Transactions from foreign adversaries and establishes a framework for rapidly responding to supply chain risk incidents. Per 15 CFR Part 7, the current prohibited list of foreign adversaries includes:

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- The People's Republic of China including the Hong Kong Special Administrative Region (China).
- Republic of Cuba (Cuba).
- Islamic Republic of Iran (Iran).
- Democratic People's Republic of Korea (North Korea).
- Russian Federation (Russia); and
- Venezuelan politician Nicolas Maduro (Maduro Regime)

ICST Transactions are defined by 15 CFR Part 7 as any acquisition, importation, transfer, installation, dealing in, or use of any information and communications technology or service, including ongoing activities, such as managed services, data transmission, software updates, repairs, or the platforming or data hosting of applications for consumer download.

- From a scorecard perspective ED will need the external system to periodically (quarterly in alignment with OSA) to provide:
 - System categorization (L/M/H)
 - Authority to Operate (ATO) dates and any triggers that would be pertinent to ED (Major upgrade, migration, etc. that changes the initial ATO status and conditions, potentially changing sys cat)
 - ATO memo provided and uploaded in CSAM
 - Does the external agency consider the system a High Value Asset
 - High level Plan of Action and Milestone (POA&M) information: Create Date, Scheduled completion date, POA&M description, Mapping to NIST SP 800-53 control(s), etc.
 - Dates and testing result for the last IRPT, CPT and DRPT
 - Level of testing/ participation needed from ED. while activating the DRP CP and IRP
 - Any CRM to show the customer responsible controls and the offered controls.
- ATO memo provided and uploaded in CSAM with the ATO dates reflected.

Per OMB-M-17-12, Preparing for and Responding to a Breach of Personally Identifiable Information, the following requirements statements should be added to all SOWs/SOOs/PWSs/BPAs/MOUs/IAAs/MOUs/ISAs that include the management of Personally Identifiable Information (PII) and / or Sensitive Personally Identifiable Information (SPII):

- The contractor must cooperate with and exchange information with agency officials, as determined necessary by the agency, in order to effectively report and manage a suspected or confirmed breach.
- The contractor and subcontractors (at any tier) must properly encrypt PII in accordance with OMB Circular A-130 and other applicable policies and to comply with any agency-specific policies for protecting PII;
- The contractor must complete regular Department training for contractors and subcontractors (at any tier) on how to identify and report a breach;

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- The contractor and subcontractors (at any tier) must report a suspected or confirmed breach in any medium or form, including paper, oral, and electronic, as soon as possible and without unreasonable delay, consistent with the agency's incident management policy and US-CERT notification guidelines;
- The contractor and subcontractors (at any tier) must maintain capabilities to determine what Federal information was or could have been accessed and by whom, construct a timeline of user activity, determine methods and techniques used to access Federal information, and identify the initial attack vector;
- The contractor must allow for an inspection, investigation, forensic analysis, and any other action necessary to ensure compliance with this Federal and Department PII Breach Response policies (such as OMB-M-17-12), the Department's breach response plan, and to assist with responding to a breach;
- The contractor must identify roles and responsibilities, in accordance with Federal and Department PII Breach Response policies (such as OMB-M-17-12), and the agency's breach response plan; and,
- The contractor must be aware that a report of a breach must not, by itself, be interpreted as evidence that the contractor or its subcontractor (at any tier) failed to provide adequate safeguards for PII.

Reporting of Data Security Breaches

If there is a suspected or known breach/disclosure of PII due to lost, theft, intercepted transfer, or other, the contractor must ensure that this breach is reported to the agency as soon as the contractor has knowledge of it. Per Office of Management and Budget Memorandum M-17-12, Federal agencies have a requirement to report breaches of PII security to the United States Computer Emergency Response Team (US-CERT).” The (PO) must notify the department within 30 minutes of discovering the incident (and the agency should not distinguish between suspected or confirmed breaches). The data security plan must be written to reflect this requirement, and the contractor must provide sufficient notification and documentation of the suspected loss, as it is understood at the time of notification to the agency for this requirement to be met. Follow-up reports of the final status of loss events will also be prepared by the contractor within a reasonable period of time as advised by the COR.

FOR E-MAIL:

In accord with BOD-18-01:

- Email Security: Agencies must configure all internet-facing mail servers to offer STARTTLS, and all second-level agency domains to have valid SPF/DMARC records. Additionally, agencies must ensure Secure Sockets Layer (SSL) v2 and SSLv3 are disabled on mail servers, and 3DES and RC4 ciphers are disabled on mail servers:
 - Within one year after issuance of this directive, issued 10-16-2017 (so, due by 10-16-2018), agencies will be required to set a DMARC policy of “reject” for all second-level domains and mail-sending hosts.

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- In accord with OMB Memorandums M-17-06, and M-15-13, and M-08-23, M-10-23, and with Binding Operational Directive (BOD) BOD-18-01, and with the NIST SP 800-52 and with the NIST SP 800-44, all e-mail applications must have SMTP enabled.

FOR NON-PUBLIC-FACING WEBSITES:

- Implement capabilities for all inbound network traffic to pass through a web content filter, which provides anti-phishing, anti-malware, and blocking of malicious websites (e.g., fake software updates, fake antivirus offers, and phishing offers)
- In accord with OMB Memorandums M-17-06, and M-15-13, and M-08-23, M-10-23, and with Binding Operational Directive (BOD) BOD-18-01, and with the NIST SP 800-52 and with the NIST SP 800-44, all Federal websites and web services must be accessible through a secure connection (HTTPS only, with HSTS), and e-mail applications must have SMTP enabled. The use of HTTPS is encouraged on intranets, but not explicitly required.
- In accord with BOD-18-01: In accord with OMB Memorandum M-08-23, in order to ensure Domain Name System Security (DNSSEC), all federal websites must be hosted on a *.gov location.

FOR PUBLIC-FACING WEBSITES:

- Implement controls to ensure that all publicly accessible externally hosted Department websites and web services only provide service through a secure connection, (such as the Hypertext Transfer Protocol Secure (HTTPS)).
- Implement controls to ensure that all publicly accessible Department websites and web services only provide service through a secure connection, (such as the Hypertext Transfer Protocol Secure (HTTPS))

In accord with BOD-18-01:

- Web Security: Agencies must ensure all publicly accessible Federal websites and web services provide service through a secure connection (HTTPS-only, with HSTS); SSLv2 and SSLv3 are disabled on web servers, and DES and RC4 ciphers are disabled on web servers; and must provide a list to DHS of agency second-level domains that can be HSTS preloaded, for which HTTPS will be enforced for all subdomains.
- If an official public-facing website will be developed, modified, or maintained, then, in accord with OMB Memorandum M-17-06, each agency must use only an approved .gov or .mil domain for its official public-facing websites. The requirement to use only approved government domains does not apply in circumstances where the agency is a user or a customer of a third-party website or service that resides on a non-governmental domain.

OMB-17-06 Policies and Requirements for Public Websites

- For requirements involving web applications, web servers, and web services, the contractor must follow the policies, principles, standards, and guidelines on information

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security and privacy, in accordance with FISMA, and implement security and privacy requirements as set forth in OMB Circular A-130 and National Institute of Standards and Technology (NIST) Special Publication 800-44, Guidelines on Securing Public Web Servers.

- The public expects Federal Government websites to be secure and their interactions with those websites to be private. The contractor must comply with requirements specified in OMB Memorandum M-15-13, Policy to Require Secure Connections across Federal Websites and Web Services, that requires that all publicly accessible Federal websites and web services only provide service through a secure connection (HTTPS with HSTS).
- The contractor must use only an approved .gov or .mil domain for official public-facing websites. The requirement to use only approved government domains does not apply in circumstances where the Department is a user or a customer of a third-party website or service that resides on a non-governmental domain. Department use of third-party websites and applications must comply with all relevant privacy protection requirements and a careful analysis of privacy implications as specified in OMB Memorandum M-10-23, Guidance for Agency Use of Third-Party Websites.
- The contractor must ensure compliance with Federal requirements to maintain public/external facing servers and services to use native IPv6. All procurements of networked information technology must comply with Federal Acquisition Regulation (FAR) requirements for use of the U.S. Government IPv6 Profile and Test Program for the completeness and quality of their IPv6 capabilities

FOR CLOUD SOLUTIONS:

- Implement controls to ensure that all publicly accessible externally hosted Department websites and web services only provide service through a secure connection, (such as the Hypertext Transfer Protocol Secure (HTTPS)).
- Only utilize FedRAMP approved cloud solutions. FedRAMP is mandatory for federal agency cloud deployments and service models at the low and moderate risk impact levels.
- If a cloud solution will be used, then an ED-issued, FedRAMP-Compliant Authorization To Operate (ATO) is a Federal and a Departmental requirement, and one must be obtained.

FOR CYBEROPERATIONS, PENETRATION TESTING, VULNERABILITY SCANNING, INTRUSION DETECTION, AND INCIDENT RESPONSE FUNCTIONS:

- Implement scanning capabilities that assess for vulnerabilities using only Security Content Automation Protocol (SCAP) validated products
- Perform penetration and regular vulnerability testing and scanning of systems, IT devices, and websites. Vulnerability scanning must be conducted at least monthly, with reports provided to the Department.
- Maintain the tools and capabilities to support asset discovery, to include passive network monitoring, active network monitoring, and automated network mapping
- Maintain tools and capabilities to support behavior monitoring, to include NetFlow analysis and network traffic capture, which captures the Transmission Control

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Protocol/Internet Protocol (TCP/IP) stream, allowing for replay of activity to determine what happened during a breach or incident

- Maintain tools and capabilities for intrusion prevention, to include host intrusion prevention systems (HIPS) and network intrusion prevention systems (NIPS)
- Maintain a firewall system designed to prevent unauthorized access to or from any contractor systems and network
- Maintain tool and capabilities to perform Security Incident/Event Management and Analysis, to include centralized logging, log correlation, and a Security Information and Event Management (SIEM) solution that provides centralized monitoring of security incidents; network behavior analytics to provide behavior-based detection to help protect against zero-day attacks; and a quarantine/sandbox environment that will isolate and analyze live traffic and/or suspected malware
- Maintain tools and capabilities to perform vulnerability and risk management and analysis to include threat intelligence threat hunt capabilities, sharing platforms, risk management or trouble ticketing system, anti-malware and anti-phishing services
- Maintain tools and capabilities to provide security situational awareness and visibility throughout the enterprise; this includes capabilities for full packet capture that collects detailed network information at the gateway and makes capture data available to analysts; endpoint incident response that enables searches all endpoints for Indicators of Compromise (IOCs) in a rapid fashion; and encrypted traffic inspection
- If working as a SOC contractor or subcontractor, will provide a Daily Morning Report, 7 days per week, that summarizes the noteworthy daily security activities. Examples include activities such as the daily count of security incidents detected (viruses, malware, etc.), email traffic analysis for spam and phishing attempts, vulnerability scan results and progress in closing open weaknesses from scan results. The contractor is encouraged to propose a world class daily security morning report format, with the most noteworthy performance measures, to include any graphic displays, and charts to enhance reporting
- Provide and maintain automated means of discovering, monitoring, and protecting sensitive data to ensure protection of data in motion, at rest, and in use
- Provide and maintain an automated means of preventing unauthorized users and computing devices from accessing contractor hosted environments, including access via remote access, wired and wireless technologies
- Provide and maintain externally facing web application firewall capabilities providing inbound and outbound traffic filtering
- Provide and maintain an Out of Band network device management solution
- Administer, operate, maintain, configure and tune cybersecurity software and hardware
- Provide full government visibility of continuous monitoring tool configurations and output on a real-time basis as well as historical data/logs
- Maintain the capability to perform periodic penetration testing, and also support for external agency red team testing, penetration testing, cyber hygiene web site vulnerability scanning tests and vulnerability assessments that the Department may need to conduct

Freedom of Information Act (FOIA) Requests

If the contractor receives a FOIA request for any data under this contract, then the contractor must immediately refer this request to the COR. NCES will process all FOIA requests related to this contract.

Implementation of Data Security Plan

The contractor must conform to the agreed-upon approved data security plan, in all activities, and must strictly enforce all procedures for ensuring confidentiality. These procedures will apply to all phases of the project, including (but not limited to):

1. Data collection in the field;
2. The transfer or transport of PII in any format, including paper records and electronic data files;
3. Coding and editing phases of data prior to machine processing; and,
4. Safeguarding and storage of master data files and response/collection documents and electronic files.

To monitor compliance with the data security plan during data collection period, the contractor must submit monthly timelines and data receipt reports that identify (1) the percent of cases transmitted within the target time; (2) the percent of cases received within the target time (paper); and (3) any outliers (e.g.: missing transmissions) and their causes with recommendations for solutions to be implemented in the next reporting period.

Physical Security for the Contractor's Facilities

The contractor's facilities must have sufficient physical security to protect restricted use information that may be kept or located there according to the procedures outlined in the NCES restricted use data manual. These requirements can be found at (<http://nces.ed.gov/statprog/rudman/>).

Security and Confidentiality Training

During the course of work on the tasks in this contract, contractor employees will be collecting, storing, editing, and otherwise handling data that are confidential. Given the restrictions on the use and handling of confidential information, all contractor employees with access to confidential information related to this contract must be required to participate in NCES approved security and confidentiality training.

Affidavit of Nondisclosure

Any contractor employee needing access to confidential information under this contract must first sign an Affidavit of Nondisclosure. Before any contractor employee starts work on the contract and has access to confidential information, a signed Affidavit of Nondisclosure must be completed. As new staff start and require access to confidential information or as staff who originally did not require such access subsequently need it, an Affidavit of Nondisclosure must be executed for them on the first working day of the assignment. The contractor must indicate the

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position in the organization of the person signing the Affidavit of Nondisclosure, and the person's functional relationship to this project in a memorandum provided to the COR.

The contractor must execute these Affidavits of Nondisclosure and the contractor must maintain the originals at its office. The contractor must make Acrobat pdf copies of new Affidavits as they are completed. Throughout the life of the contract, the memorandum and Affidavits of Nondisclosure for new project staff, as well as interviewers and other short-term personnel, will be submitted on a schedule designated by the COR or, at a minimum, three times a year. The contractor must be able to produce the original hard copies of the Affidavits within a few hours' notice from the COR. The NCES data security officer can provide blank copies of the Affidavits upon request.

Quality Control Measures

NCES is firmly committed to determining how well each project meets its goals, the quality of data collected, and its implications for future projects. Project quality control must be integrated into every phase of the project. For instance, potential problems should be identified early so that possible solutions are tested as part of the field trial. Item reliability and validity tests must be built into the data collection efforts, non-response bias analyses should be conducted as part of the final weighting process, and survey methodology and sample design effects should be made available to all data users. Thus, statistical quality control must be incorporated into each project stage as appropriate.

Property of Information Technology Programming or Source Code

When programming and or source code is used to develop a deliverable, the contractor will be required to provide the annotated code when the deliverable is submitted to and accepted by the Government. All intermediate and final products and supporting programming and or source code generated by work under awarded tasks are the property of the Government, and must be provided to the Government upon request. Analyses and data developed under this contract by contractors, subcontractors, and or their partners cannot be released without written NCES approval. Contractors, subcontractors, and or their partners may not use pre-release or restricted-use data provided under this contract for any purpose not specified in the contract without written permission from the COR and Task Leader.

Appendix C. Disclosure Analysis Plan

Before data files can be made public or submitted to the OECD, the contractor must prepare and implement a Disclosure Analysis Plan (DAP). The DAP should identify any variables at high risk for identifying participants in the study and should provide a proposal for how to maintain the confidentiality of these respondents through data perturbation. The DAP will include proposals for data perturbation which will follow NCES standards and will necessitate data matching (to prevent an intruder from matching to external data to re-identify respondents), swapping, and other methods of disclosure limitation. The checklist below provides a brief overview of what will be expected of the contractor in preparing and implementing the DAP. More explicit details on the DAP will be provided by the COR at a later date.

Information checklist for disclosure analysis plans

This checklist provides guidance for writing a disclosure analysis plan (DAP) for a microdata file. This particular checklist is modeled on DAPs sent to and approved by the IES disclosure review board (DRB). The length of each DAP section depends on the survey and any specific DRB requirements for the study.

1. Executive Summary

2. Introduction

- ___ State the data release modes and their interrelationship.
- ___ State purpose of the survey.
- ___ State the various samples selected that the plan covers.
- ___ Summarize the steps to reduce disclosure risk.
- ___ Summarize the initial risk analysis.
- ___ Mention compliance with agency standards.
- ___ Mention compliance with the study's contractual agreement.
- ___ Summarize how the data are planned for release.

3. Sample Design, Weighting, and Variance Estimation

- ___ List the stages of sampling.
- ___ Briefly explain how sampling units were selected at each stage.
- ___ Provide a summary of sample sizes for each sampling stage.
- ___ Discuss any domains with differential sampling rates (oversampling procedures).
- ___ Briefly describe the weighting procedures.
- ___ Describe any special variance estimation procedures.

4. Results of Initial Risk Analysis

(This section describes the initial risk analyses that should be completed prior to the DAP.)

- 4.1 Direct Identifiers - Respondent Identifiers and Geographic Identifiers (to be deleted from data):
 - ___ List the key variables that could be used for unit-level identification (e.g. person, school, establishment).
 - ___ List the key variables that could be used for geographic identification.

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- ☐ List the variables used for variance estimation.
 - ☐ List any contextual variables, that is, those that can indirectly identify a geographic area such as percentages of minority groups, or other population characteristics.
 - ☐ List any cases with outlying weight values.
 - ☐ List any other variables that identify sample units as unique in the population or unique in the sample.
 - ☐ List variables that could identify groupings of sampled cases.
 - ☐ Indicate variables that may contain extreme values.
- 4.2 Ability to Match with Administrative Data and Other External Data:
- ☐ List publicly/commercially available data sources (including the frame) that could be used to identify sampled units.
 - ☐ Describe the proposed approach and software used to match the masked file against the publicly available files.
 - ☐ Describe the data modification needed due to the matching outcome
- 4.3 Questionnaire Data Items with Nonnegligible Risk:
- ☐ List questionnaire items or group of items that may put confidentiality at risk.
 - ☐ Classify these questionnaire items as factual or subjective to determine their risk level for disclosure.²
 - ☐ Report results of frequencies on factual variables crossed with key variables to determine disclosure risk.
- 4.4 Discussion of disclosure risk from weights or variance estimation variables
- 5. Data Coarsening for Disclosure Protection**
- 5.1 Variable Suppression:
- ☐ List the key variables that need suppression.
 - ☐ List the factual variables that need suppression.
- 5.2 Variable Recoding:
- ☐ List the variables that need categorization or recategorization, and the new categories.
 - ☐ List of variables that need top- and/or bottom-coding and the new categories.
- 6. Data Swapping for Disclosure Protection**
- ☐ Briefly explain the purpose of data swapping.³
 - ☐ Describe the steps of the procedure.
 - ☐ State the software to be used to perform the procedure.
 - ☐ List and describe all variables involved, as appropriate.

² The disclosure risk analysis does not generally apply to items collecting subjective information, since data intruders can never be sure of the identity of a sampled unit based on responses to a subjective item.

³ Data swapping is required for IES projects. If a program office decides not to implement data swapping, then a justification for no swapping must be provided.

ATTACHMENT 1 – PERFORMANCE WORK STATEMENT

— Discuss special steps needed if highly visible cases exist.

7. Evaluating the safe file

— Discuss and show the impact of swapping procedures on data quality and how it will be measured.

— Describe the degree of the consistency between the analytic results obtained from the safe file and other data releases with the same information (DAS, static tables, a restricted use microdata file (RUMF), etc.).

Appendix

List all variables and whether they are suppressed or recoded.

Appendix D. Excerpts from 2025 PISA Computer Based Assessment Preparation Manual (Version 1 – May 2023)

1 Computer-Based Testing in PISA 2025

1.1 Internet browsers

PISA 2025 CBA Online can be used on four common internet browsers. For each internet browser there are requirements about which version is used – all of these internet browsers can easily be updated on school computers.

1.2 Computer requirements

Students are not able to complete PISA 2025 testing using a mobile device such as a smart phone or a tablet

For PISA 2025 each student needs to use a device with the specifications identified below.

1.2.1 Type of device

Three types of devices can be used:

- computer
- laptop, or
- netbook.

1.2.2 Screen size and resolution

Devices should have the following:

- screen size of at least 24.6 cm (9.5 inches) diagonally
- resolution of at least 1280 x 720 pixels

1.3 Headset requirements

In countries that have chosen to participate in the Foreign Language Assessment (FLA) in PISA 2025, all students require a headset. Specifications can be found in the FLA School Logistics Manual.

Schools may need to install headset drivers on computers in advance of testing to ensure time for troubleshooting. Schools also need to check that there are sufficient ports on each computer to connect both a mouse and a headset at simultaneously if required.

2 School Readiness Tool

The PISA 2025 School Readiness Tool has been developed to assist schools identify whether they can proceed with CBA Online or need to use CBA Offline instead. The School Readiness Tool will identify if a schools' internet bandwidth will be sufficient is sufficient for CBA Online.

The School Readiness Tool is extremely easy for schools to use, following these three steps:

- 1) Paste the hyperlink into an internet browser
- 2) Indicate if the country is implementing the Foreign Language Assessment (FLA)
- 3) Wait for the results

The PISA 2025 School Readiness Tool User Guide provides instructions about how to use this tool.

3 CBA Online

3.1 Preparing for CBA online

Most students will participate in PISA 2025 using CBA Online. Many schools have participated in the PISA computer-based assessment in previous cycles. This means that they will be familiar with PISA testing procedures used in the past.

With the change to a new test delivery platform in PISA 2025 it is important that National Centres become aware of what is different and how they will implement the new test delivery platform in their country/economy.

This section provides details that National Centres can use to help them understand what will be required and to assist them to prepare accordingly.

It is important to note that this manual is for preparation only – it does not include instructions for implementation. Future manuals will be provided that will give the detailed information that schools, and test administrators will need to implement in the field trial, and eventually the main survey. These include:

- PISA 2025 Field Operations Manual
- PISA 2025 School Coordination Manual
- PISA 2025 School Associate Manual
- PISA 2025 Test Administration Manual
- PISA 2025 Test-taker Delivery System Manual

3.2 Steps in preparing schools for CBA Online in PISA 2025

To help National Centres prepare schools for CBA Online in PISA 2025, Figure 1 provides a summary of steps to go through with all sampled schools (SRT = School Readiness Tool).

ATTACHMENT 1 – PERFORMANCE WORK STATEMENT

For schools that have participated in PISA before, it is particularly important to emphasise how CBA Online is different from the CBA in previous PISA cycles.

Figure 1: Preparing school readiness for PISA 2025

Inform sampled schools about CBA options and requirements for PISA 2025, and whether your country or system is participating in the FLA

1. Use the SRT User Guide to assist schools to use the SRT
2. For schools with a successful SRT result - inform them about CBA Online requirements
3. For schools with an unsuccessful SRT result - encourage them to use the solutions listed in the SRT User Guide
4. For schools with a successful SRT result after using solutions - inform them about CBA Online requirements
5. For schools with an unsuccessful SRT result despite using solutions - inform them about CBA Offline requirements
6. Use resources provided by the International Consortium to inform and train School Coordinators, IT Coordinators and/or School Associates on preparing for testing
7. Prepare the Test Administration and/or School Associate Manuals for CBA Online and CBA Offline based on the manuals provided by the International Consortium

3.3 CBA Online assessment requirements

To participate in PISA 2025 via CBA Online students need only three things:

1. A computer, laptop or netbook that meets the technical requirements outlined in section 4.
2. A connection to the internet (wireless or wired) that has passed the School Readiness Test.
3. A username and password combination.

If schools are located in a country that is participating in the Foreign Language Assessment (FLA), then a headset that meets the technical requirements defined in the Foreign Language Assessment School Logistics Manual.

3.4 Setting up student computers for CBA Online

Each student requires a desktop computer, laptop or notebook with internet access to participate in the CBA Online.

The login address will be a simple website address.

It is important for schools to make sure that the webpage is not blocked by school security settings.

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This must be done IN ADVANCE of the date of testing to enable sufficient time for the webpage to be unblocked.

3.5 Student logins

Students will need usernames and passwords to access the system.

Each school will have either a School Associate (SA) (a member of the school staff) or a Test Administrator (TA) (an external staff member appointed by the National Centre) to administer the assessment.

National Centres will prepare the Student Login Forms, which include the login and password for each sampled student and provide these forms to the SA or TA. The SA or TA is responsible for providing each sampled student with their corresponding Student Login Form.

The SA or TA will read a script aloud to students. This will include instructions for the assessment session. When told to do so, to login to system, students simply enter their login and password.

3.6 Practice assessment

Before students start the test session, they will be able to undertake some practice questions on the Test Delivery System. The TA or SA will provide a set period of time for students to do this. There will be practice items for the different item formats that students will be presented with when they are completing the assessment. These include:

- a simple multiple-choice question (where students select one option)
- a complex multiple-choice question (where students select more than one option)
- a drag and drop item (where students drag and drop different elements)
- a complex multiple-choice question (where students select an option and then type a reason)
- a constructed response task (where students provide a short, typed response)
- a simulation item
- a questionnaire item

3.7 Navigating through the test session

At the end of the practice test session, students will encounter a ‘holding screen’ (HOLD). At this point, the SA or TA will read aloud more instructions.

As they move through the assessment session there will be a number of holding screens.

At each of the steps marked HOLD, the TA or SA will provide verbal instructions to students and then give them another password to enter.

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The password to start each section will be a simple four-digit password e.g., 1234 or 0000.

3.8 Features of the assessment screen

The PISA 2025 Test Delivery System is designed to be very user-friendly. Most screens show the same set of elements.

Students are easily able to track their progress, navigate forwards and backwards (within units), view the units they have completed and advance through the assessment. They can also see how much time they have left for each section.

Student responses are automatically saved as they move through the test. If they have to stop the test – for example due to an internet disruption, an emergency or student illness, they can log in using the same user details and the test session will restart at exactly the same point.

Features such as a calculator will only appear if required in a unit.

Interactive items will appear somewhat different to non-interactive items, but the same screen features will remain visible.

3.9 Multi-stage adaptive assessment

The PISA 2025 assessment is a multi-stage adaptive assessment (MSAT). In the Main Survey this means that:

- if students perform well on earlier units, the test system will present them with more difficult units; and
- if students perform poorly on earlier units, the test system will present them with easier units.

Each element of the assessment is called a ‘Testlet’. Each Testlet includes a number of assessment units, with several items within each unit.

- Testlet – a cluster of a number of units
- Unit – a group of items on a particular topic
- Item – individual question or activity within a unit

The objective of using MSAT is to have more precise measurement of student proficiency than is possible with non-adaptive testing.

This is designed to be seamless for students. This means that students will not know which pathway they have taken through the test.

3.9.1 Field Trial design for MSAT

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For the Field Trial, the clusters of units will be ordered in many different ways, including new science items (S), Learning in the Digital World (LDW), and the Foreign Language Assessment (FLA).

Students will take two 60-minute test blocks. For Science (S) and Learning in the Digital World (LDW), each block will include two clusters. The goal is to collect sufficient data on each item and unit to enable psychometric analysis after the Field Trial.

Since mathematics and reading items are all trend items that have been used in one or more previous cycles of PISA, there is no need to include them in the Field Trial.

3.10 Checking progress

Students can review which items they have completed within the assessment by selecting the ‘OVERVIEW’ button. This will take them to a summary screen where they can track their progress.

3.11 Completing the effort thermometer and questionnaire

After students complete the assessment, they will see a holding screen. The SA or TA will read further instructions and students will then complete the effort thermometer and questionnaire.

In some schools, students may complete the questionnaire in a different test session.

3.12 Ending the test session

When the time for the questionnaire ends, the test delivery system will automatically change to a final screen. This will confirm that the student has completed both the assessment and questionnaire. This will be similar to the image shown in Figure 8.

The TA or SA will provide instructions to students to end the test and students will leave the assessment room.

3.13 Resuming the assessment session

If it is necessary for a student to interrupt an assessment session at any point, for example, due to illness, connection issues, or an emergency, the PISA 2025 Test Delivery System will keep a record of their precise location in the test or questionnaire.

To resume, the student simply re-enters their username and password. The test session will start at the same location that the previous session ended.

4 IT capacity evaluation

National Centres are required to evaluate their IT capacity to ascertain how PISA will be delivered in their country/economy.² This will involve the following steps:

- 1) Test the online capacity using the School Readiness Tool (Phase 1)

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- 2) If required, test the offline capacity using the Offline Demo Tool (Phase 2)
- 3) Provide your country's preferred PISA delivery method using the Confirmation of Delivery Mode Form (Phase 3).

4.1 Phase 1: Testing your country's capacity to deliver CBA Online

The National Centre will need to select a number of schools that reflect the diversity of the IT resources in schools and the different kinds of schools (for e.g., inner city schools, remote schools) across the country to assess if the assessment can be delivered online. For example:

- If schools in your country have the same IT devices:
 - randomly select a number of schools, for e.g., at least 15 schools, (i.e., one-tenth of the number of schools in the Main Survey school sample)
 - ask these schools to run the School Readiness Tool, using the instructions in the School Readiness Tool User Guide, at a time(s) that will be comparable to the period when students will be completing the PISA assessment
 - ask these schools to provide the results from running the School Readiness Tool.
- If schools in your country have different IT devices (for e.g., some schools may have Windows desktop computers, and other schools may have a combination of Windows and Mac laptops), then your National Centre will need to select as many schools as necessary to evaluate if the PISA can be delivered online.

Once schools have returned the results from running the School Readiness Tool³, your National Centre should analyse the results to determine the school types that can run PISA online and the school types that might need an offline solution.

4.2 Phase 2: Testing your country's capacity to deliver CBA Offline

For those school types who appear to not be able to successfully run the School Readiness Tool, your National Centre will need to:

- ask these schools to run the Offline Demo Tool
- ask these schools to provide the results from running the Offline Demo Tool.

4.3 Phase 3: Final IT capacity analyses and consultation with ACER

Once your National Centre has completed its evaluation of running the PISA assessment in schools, using either the online and/or offline delivery methods, you will need to:

1. Complete the Confirmation of Delivery Mode Form (see Appendix A)
2. Upload the complete the Confirmation of Delivery Mode Form to the 'General' folder in the File Exchange on PISA Connect.

ATTACHMENT 1 – PERFORMANCE WORK STATEMENT

3. Send a message via Help desk to advise that the Confirmation of Delivery Mode Form has been uploaded to the File Exchange. If you still have concerns about the delivery of PISA in your country/economy, request a consultation with ACER to discuss these matters further.

Please complete this task by now later than Tuesday 31 October 2023.

5 Finding out more about CBA Online and Offline

The purpose of this manual is to provide an overview of the approach to assessment for CBA Online and Offline, and instructions for evaluating how your country/economy will deliver PISA.

5.1 Detailed manuals

The International Contractor will provide detailed manuals including step-by-step instructions on how to implement testing in both CBA Online and CBA Offline.

The manuals will include all technical details for preparing computers in advance of testing. They will also include scripts for TA or SA to read aloud during the testing sessions.

5.2 Help and support

The International Contractor is available to provide support in a timely manner.

- PISA Connect Help Desk - Before the Field Trial, and between the Field Trial and Main Study, National Centres can submit tickets to the PISA Connect Help Desk.
- Technical Support - During the Field Trial and Main Survey, the International Contractor will be available for technical support via the PISA Connect Help Desk. Help Desk will provide 16-hour a day, 5 day a week support for National Centres during these critical testing windows.

Appendix E. PISA 2025 School Readiness Tool User Guide

PISA 2025 will be delivered as an online assessment via a browser-based application.

It is important to establish whether the computers and laptops that students will use to administer PISA can successfully run the assessment.

The School Readiness Tool (SRT) assesses the compatibility of the schools' IT resources with the assessment platform and identifies if they perform at the appropriate level.

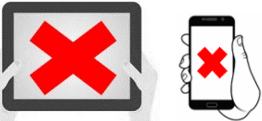
This user guide provides details about the technical requirements and the instructions for running the SRT.

Purpose of the SRT

The School Readiness Tool is intended for schools to assess their schools' IT capacity to deliver PISA online.

Device requirements

The device requirements for the assessment platform are:

A computer, laptop, or netbook	
	Specifications: - screen size at least 24.6 cm (9.5 inches) diagonally - resolution at least 1366 x 768 pixels
Other requirements	
Physical keyboard (wired or Wi-Fi)	
Mouse or trackpad	
Not permissible	
Students are not able to complete the PISA 2025 assessment using a tablet, touchscreen, or mobile device such as a smart phone.	

Additional requirements for the Foreign Language Assessment (students will need to complete listening and speaking tasks):

Component	Essential	Avoid
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ATTACHMENT 1 – PERFORMANCE WORK STATEMENT

Headphones	 • Over-ear ○ Sitting on the head, ○ encircling and covering the full ear ○ Good padding for comfort • Closed back ○ Ear cups have no air gaps or airflow to the ear Note: Headphones do not need to be noise-cancelling.	 • On-ear headphones  • In-ear or earbud headphones
Microphone	• Boom directional microphone • Attached to the headset • Noise-cancelling (to maximise student pick-up and to minimise noise pollution on the recordings)	• Built-in microphones (as they will pick up all surrounding noises and are not appropriate for the assessment environment) • Computer microphones • Hidden microphones • Student mobile phones
Connection to the computer	• Wired connection via a stereo 3.5mm audio jack or USB-A connector • Cord length that allows students to wear the headset comfortably	• Bluetooth connections should not be used because of the negative impact on sound quality

Notes: Headphones may require appropriate drivers to be installed and confirmed as working before the test day.

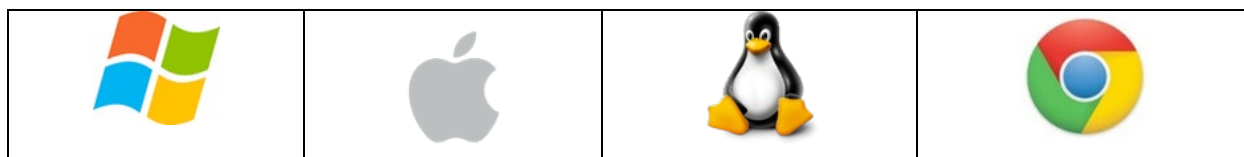
All built-in microphones on computers should be disabled.

System requirements

TAO® can be used on standard desktop or laptop computers with a standard desktop edition of the Windows, macOS, Linux, or Chrome operating system (OS) installed.

Operating Systems			
Windows	macOS	Linux	ChromeOS

ATTACHMENT 1 – PERFORMANCE WORK STATEMENT



PISA 2025 CBA Online can be used on four common internet browsers. For each internet browser there are requirements about which version is used – all of these internet browsers can easily be updated on school computers.

Important: The Operating System chosen must meet the browser requirements listed here and the version of the chosen browser must not be more than 12 months old.

Internet Browsers			
Google Chrome	Mozilla Firefox	Microsoft	Apple Safari
			

Internet requirements

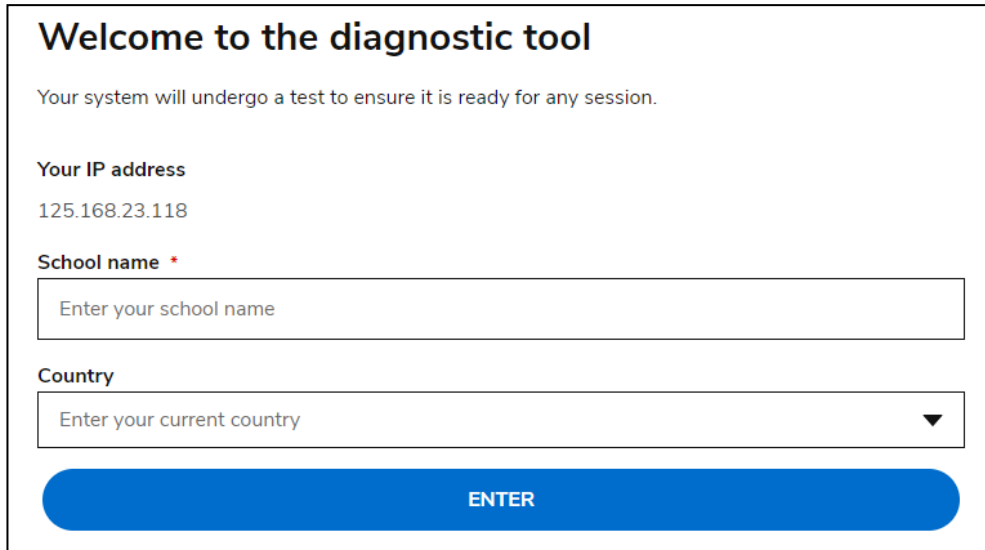
The internet requirements to access the assessment platform for **50 concurrent users** are:

- An internet connection (broadband wired or wireless) with:
 - at least 4MB per second (40 Mbps) download
 - at least 1.2MB per second (12Mbps) upload
 - maximum acceptable latency – 800ms

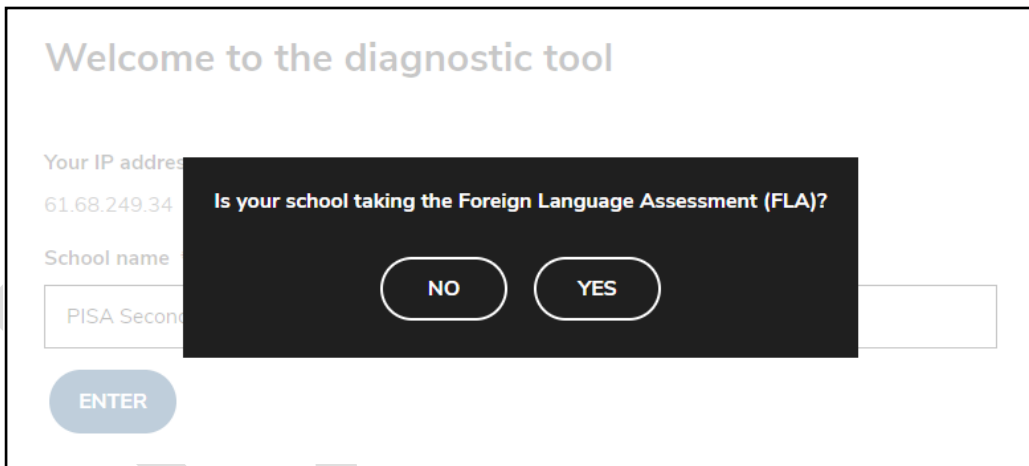
Note: The SRT has been configured for 50 concurrent users. For less than 50 users please refer to the chart in the in bullet point 6 (non-FLA) and point 10 (FLA) under section 6.

Running the School Readiness Tool

1. Go to: <https://schoolreadiness-qa.pisa.acer.org/>
2. After clicking the link, the SRT landing page will appear:



3. Type in your school name (It can be a dummy name, it doesn't need to be the real name of the school) and select your country, then click *Enter*.
4. After you click *Enter*, the prompt for FLA will appear:



If your country/education system will not be participating in the FLA à click *NO*.

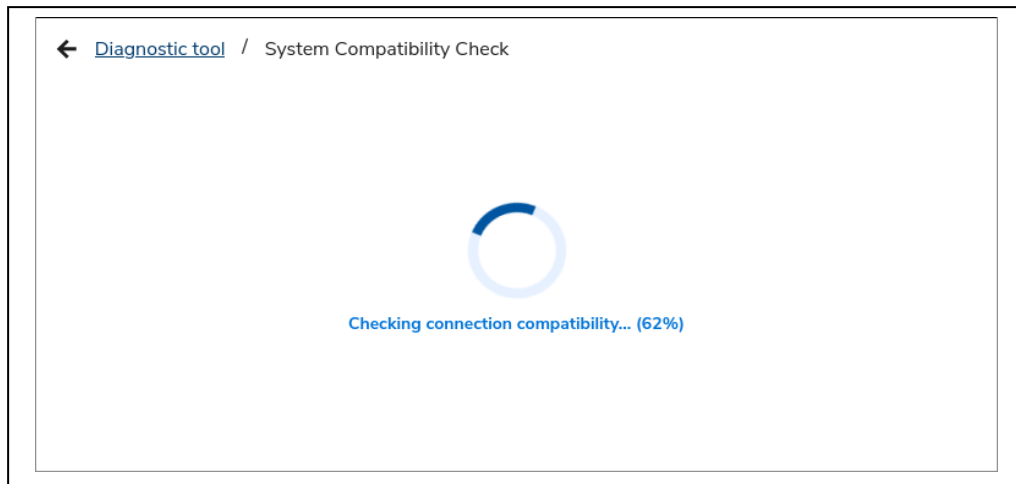
- The SRT will run a system check, please refer to steps 5 and 6 for instructions.

If your country/education system will be participating in the FLA à click *YES*.

- The SRT will run an audio and a system check, please refer to steps 7 to 10 for instructions.

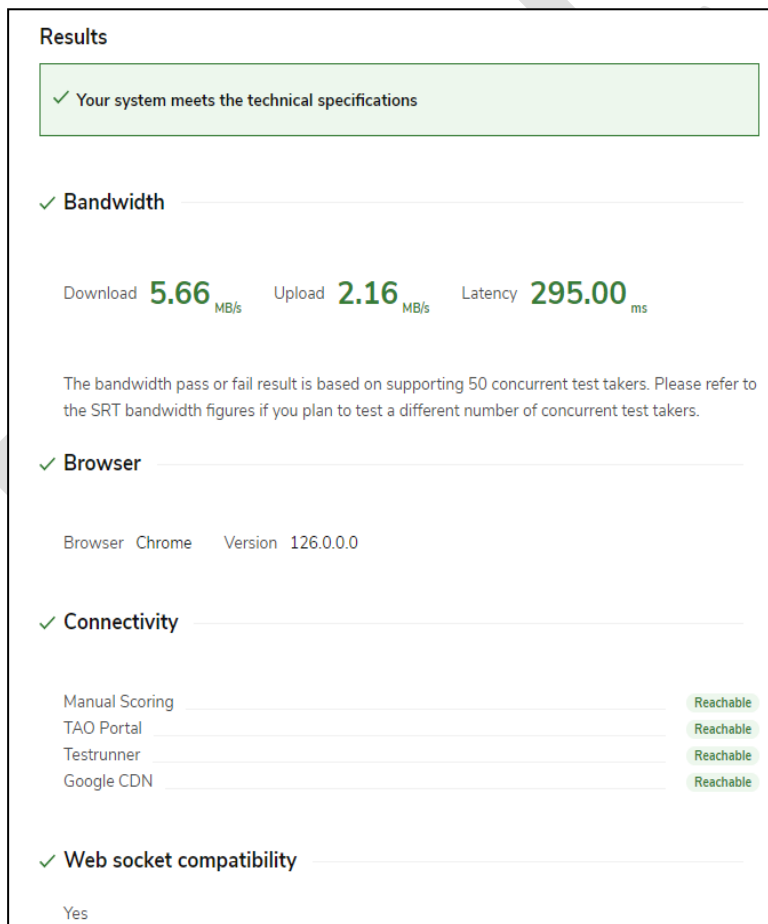
5. After you click *NO*, the SRT will perform system compatibility checks. This could take a few minutes. The screen will show you the progress of running these checks.

ATTACHMENT 1 – PERFORMANCE WORK STATEMENT



6. Once the system compatibility checks have been completed, the results will be displayed on the screen.

The following will be displayed if the system meets the technical specifications:



The following screen will be displayed if the system does not meet the technical specifications **based on 50 concurrent test takers**:

ATTACHMENT 1 – PERFORMANCE WORK STATEMENT

Results

⚡ Your system does not meet the technical specifications and is not suited to run this assessment.

⚡ Bandwidth

Download **5.74** MB/s Upload **267.59** KB/s Latency **336.00** ms

The bandwidth pass or fail result is based on supporting 50 concurrent test takers. Please refer to the SRT bandwidth figures if you plan to test a different number of concurrent test takers.

✓ Browser

Browser Edge Version 127.0.0.0

✓ Connectivity

Manual Scoring	Reachable
TAO Portal	Reachable
Testrunner	Reachable
Google CDN	Reachable

✓ Web socket compatibility

Yes

The below graph provides a guidance if you plan to test a different number of concurrent test takers.

Figures 1 and 2 show the estimated bandwidth needed for uploading and downloading by the number of concurrent users. The lines are not “go/no go” criteria, they are estimates and approximate.

The National Centre will need to review schools near the line and decide if there is enough evidence to support that these schools can administer PISA online or not; for example, if they have conducted online assessments before or if they use computers/laptops and the internet often.

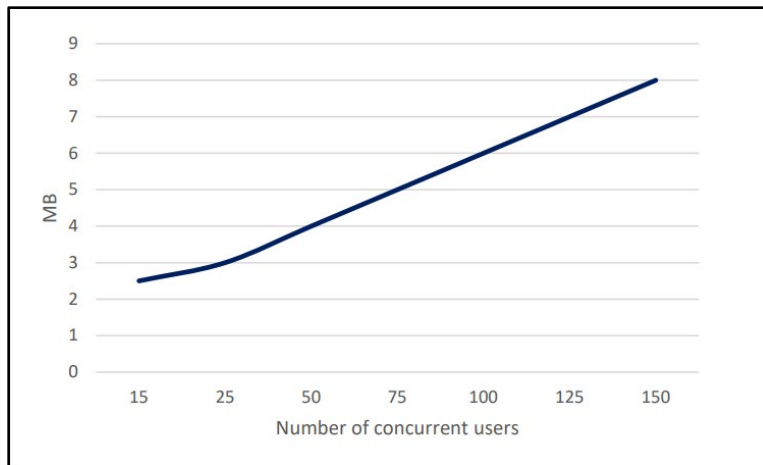


Figure 1 Recommended download speed

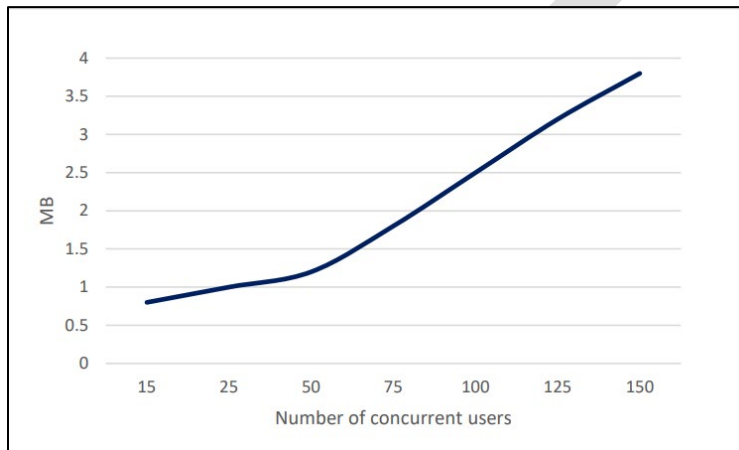


Figure 2 Recommended upload speed

Connectivity

Whitelisting a domain is a security measure where specific web addresses (domains) are explicitly allowed to communicate with a network or application. Whitelisting ensures that traffic to and from specific web addresses is permitted within the network's security settings, allowing users to access essential resources without being blocked by firewalls or content filters.

In the SRT test, we are checking the connection to all domains needed:

- Manual Scoring
- TAO Portal
- TestRunner
- Google CDN

ATTACHMENT 1 – PERFORMANCE WORK STATEMENT

To make sure connection to above domains work IT personnel need to make sure below URLs are whitelisted

Online:

pisa2025.org/locale/

prod.gcp-eu.taocloud.org/

pisa2025-portal.prod.gcp-eu.taocloud.org/

auth-pisa2025.prod.gcp-eu.taocloud.org/

testrunner-pisa2025.prod.gcp-eu.taocloud.org/

Offline:

pisa2025.local/portal/locale

prod.gcp-eu.taocloud.org/

pisa2025-portal.prod.gcp-eu.taocloud.org/

auth-pisa2025.prod.gcp-eu.taocloud.org/

testrunner-pisa2025.prod.gcp-eu.taocloud.org/

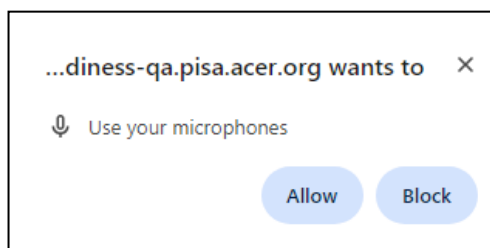
Web socket Compatibility

WebSockets reduce latency and improve data exchange efficiency by maintaining an open connection, allowing both parties to send and receive data immediately as needed.

By thoroughly verifying both WebSocket compatibility and domain whitelisting, schools can ensure that their network infrastructure supports real-time to allow seamless user experience.

7. After you click *YES*, the audio check screen will appear.

Note: A browser prompt may be displayed requesting access to your microphone. Click *Allow*. This will enable the audio check to commence. (In the event you click *Block* by mistake, please restart the SRT by clicking on the link.



ATTACHMENT 1 – PERFORMANCE WORK STATEMENT

This audio check screen shows the:

- **microphones** available on your device. The default microphone will be selected. You can use the default microphone, or you can select another microphone. **Please test the microphone that will be used by students during the FLA.**
- **Input level** indicates the level of audio into the selected microphone.
- **Output level** indicates the level of audio from the speaker.

Audio check

Select a microphone

☐ Communications - Microphone (Conexant ISST Audio)

☒ Microphone (Conexant ISST Audio)

Input level

Output level

To test your microphone and speaker please begin by recording a short audio. Once complete you can playback your recording.

START RECORDING

CONTINUE TO SYSTEM CHECK →

Click the *START RECORDING* box and speak into the microphone. After clicking the *START RECORDING* box, the 'Recording...' box will appear, as shown below. This indicates you can say something to test the audio capacity. For example, you could say 'Testing, 1, 2, 3, testing, 1, 2, 3'.

STOP RECORDING

Recording...

Click the *STOP RECORDING* to stop the audio recording.

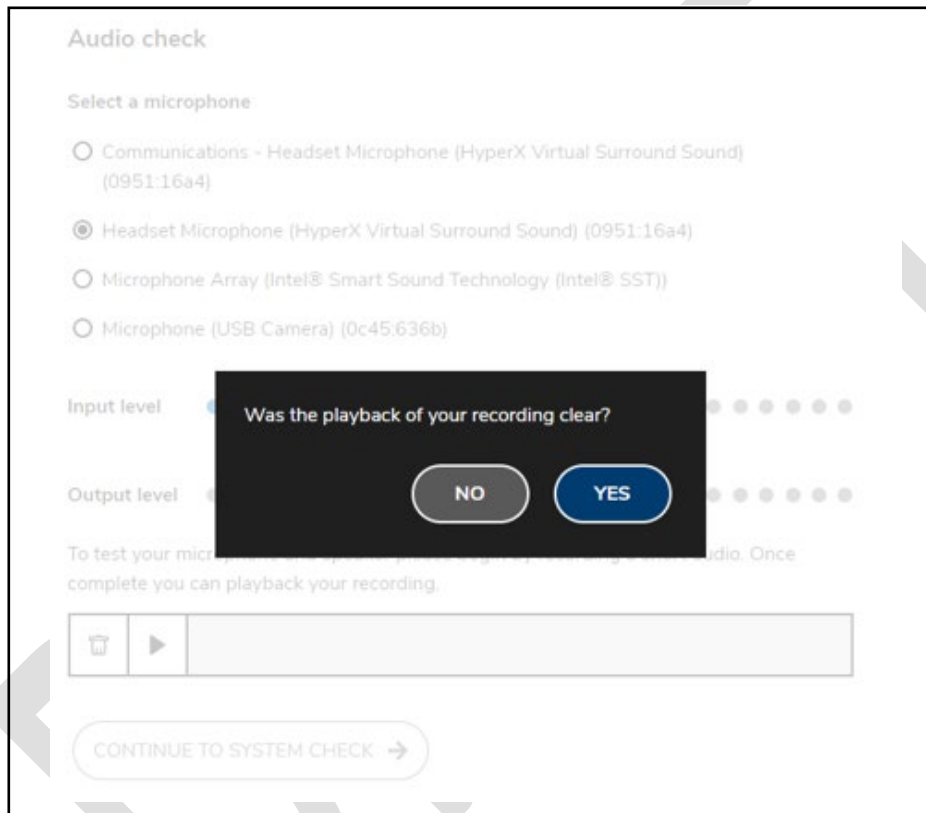
Click the play (▶) button to hear your recording. You can pause the playback by selecting the pause (⏸) icon.

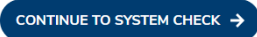
ATTACHMENT 1 – PERFORMANCE WORK STATEMENT

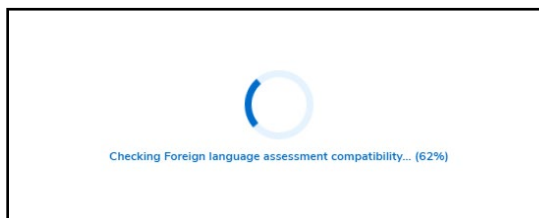


8. After you have listened to the playback of your recording, you will be prompted for a response on the clarity of the recording. Click:
- *NO*, if you cannot hear your recording clearly
 - *YES*, if you can hear your recording clearly.

This response will be reported in the results page.



9. After providing your response, click the  button to move on to the system compatibility checks. This could take a few minutes. The screen will show you the progress for running these checks.



ATTACHMENT 1 – PERFORMANCE WORK STATEMENT

10. Once the system compatibility checks have been completed, the results will be displayed on the screen.

The following screen will be displayed if the device meets the system and audio specifications.

(The *Playback* will be *YES* if you selected *YES* for the audio clarity question.)

Results

✓ Your system meets the technical specifications

✓ **Bandwidth**

Download **5.61** MB/s Upload **2.18** MB/s Latency **292.00** ms

The bandwidth pass or fail result is based on supporting 50 concurrent test takers. Please refer to the SRT bandwidth figures if you plan to test a different number of concurrent test takers.

✓ **Browser**

Browser Chrome Version 126.0.0.0

✓ **Connectivity**

Manual Scoring	Reachable
TAO Portal	Reachable
Testrunner	Reachable
Google CDN	Reachable

✓ **Web socket compatibility**

Yes

✓ **Audio**

Playback Yes

The following screen will be displayed if the device does not meet the system and audio specifications based on 50 concurrent test takers. (The *Playback* will be *NO* if you selected *NO* for the audio clarity question.)

ATTACHMENT 1 – PERFORMANCE WORK STATEMENT

Results

⚡ Your system does not meet the technical specifications and is not suited to run this assessment.

⚡ Bandwidth

Download **3.51** MB/s Upload **351.91** KB/s Latency **343.00** ms

The bandwidth pass or fail result is based on supporting 50 concurrent test takers. Please refer to the SRT bandwidth figures if you plan to test a different number of concurrent test takers.

✓ Browser

Browser Edge Version 127.0.0.0

✓ Connectivity

Manual Scoring	Reachable
TAO Portal	Reachable
Testrunner	Reachable
Google CDN	Reachable

✓ Web socket compatibility

Yes

✓ Audio

Playback Yes

[RETEST SYSTEM](#)

The below graph provides a guidance if you plan to test a different number of concurrent test takers.

Figures 1 and 2 show the estimated bandwidth needed for uploading and downloading by the number of concurrent users. The lines are not “go/no go” criteria, they are estimates and approximate.

The National Centre will need to review schools near the line and decide if there is enough evidence to support that these schools can administer PISA online or not; for example, if they have conducted online assessments before or if they use computers/laptops and the internet often.

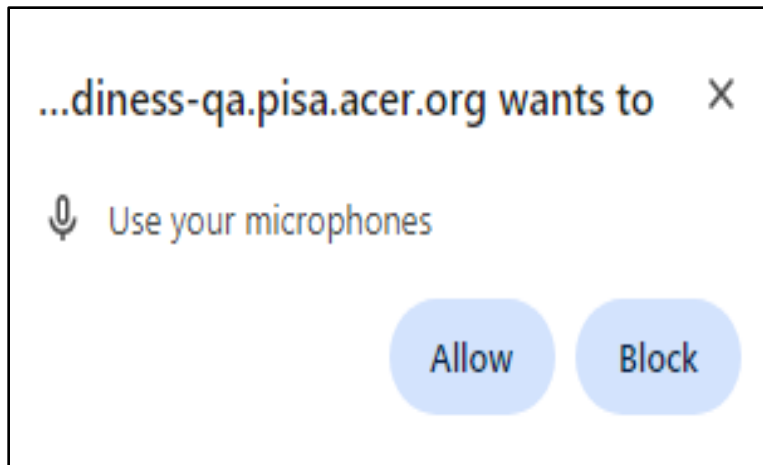


Figure 1 Recommended download speed

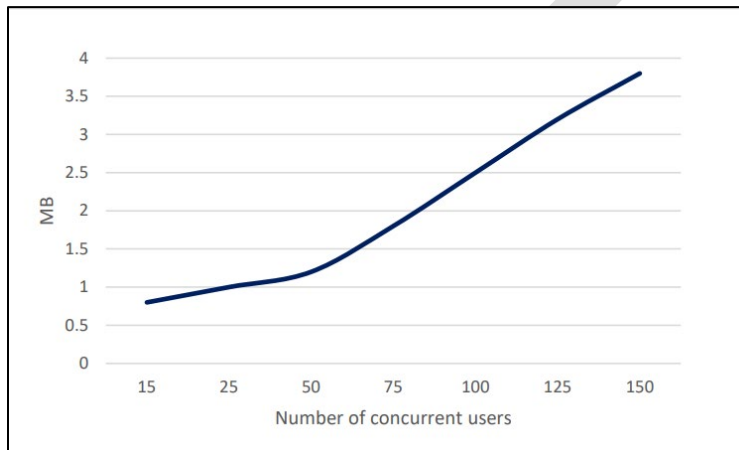


Figure 2 Recommended upload speed

Connectivity

Whitelisting a domain is a security measure where specific web addresses (domains) are explicitly allowed to communicate with a network or application.

Whitelisting ensures that traffic to and from specific web addresses is permitted within the network's security settings, allowing users to access essential resources without being blocked by firewalls or content filters.

In the SRT test, we are checking the connection to all domains needed:

- Manual Scoring
- TAO Portal
- TestRunner
- Google CDN

ATTACHMENT 1 – PERFORMANCE WORK STATEMENT

To make sure connection to above domains work IT personnel need to make sure below URLs are whitelisted

Online:

pisa2025.org/locale/

prod.gcp-eu.taocloud.org/

pisa2025-portal.prod.gcp-eu.taocloud.org/

auth-pisa2025.prod.gcp-eu.taocloud.org/

testrunner-pisa2025.prod.gcp-eu.taocloud.org/

Offline:

pisa2025.local/portal/locale

prod.gcp-eu.taocloud.org/

pisa2025-portal.prod.gcp-eu.taocloud.org/

auth-pisa2025.prod.gcp-eu.taocloud.org/

testrunner-pisa2025.prod.gcp-eu.taocloud.org/

Web socket Compatibility

WebSockets reduce latency and improve data exchange efficiency by maintaining an open connection, allowing both parties to send and receive data immediately as needed.

By thoroughly verifying both WebSocket compatibility and domain whitelisting, schools can ensure that their network infrastructure supports real-time to allow seamless user experience.

The SRT is the tool schools will use to undertake the technical preparation activities to determine the most effective way to deliver PISA. For these reasons, it is important to run the SRT several times. Considering the computers/laptops students will use to participate in PISA, the SRT should be run:

- on all computers/laptops with different configurations
- using different browsers
- at different times of the school day and the school week.

Results

After running the SRT several times, you will need to review the results from running the SRT and provide your results to the <National Centre> using the *SRT Results Form* (see Appendix A).

If a device has met all the system specifications and audio specifications (for FLA Participants), then the PISA assessment can be delivered online.

If a device does not meet all the system specifications and audio specifications (for FLA Participants), then this requires further investigation (see Section 8 Troubleshooting).

Troubleshooting

As a starting point, check the usage of the platform to determine usage levels, check the equipment, including the headset/microphone connection (for FLA Participants), reset the network and run the SRT again.

Issue	Action
Internet connectivity is poor OR latency has failed	Please ensure you are running the SRT in a timeframe that will be comparable to the period when students will be completing the PISA assessment. If the problem persists, please contact your internet provider.
Browser has failed	Please update your browser or install a supported browser from the list provided under Section 4. System Requirements
Internet network conflict (such as having wifi enabled, when the student computers/laptops are physically connected to the school network)	Please ensure you are using the correct network when running the SRT.
For FLA participants, if the audio has failed	The <National Centre> will be required to assess the audio equipment. If the audio equipment continues to fail the audio specifications, headphones and microphones will need to be replaced.
WebSocket compatibility fails	There are instances where WebSocket connections fail to establish successfully, leading to WebSocket errors like interrupted data transmission and compromised functionality. Here are some common reasons for WebSocket connection failures and their corresponding solutions: Network Issues: Unstable or unreliable network connections can lead to WebSocket connection failures. This may include issues such as high latency, packet loss, or intermittent connectivity problems.

	Solution: • Verify your network connection and ensure internet access on your device • Check for firewalls or proxy server restrictions on WebSocket connections • Test with different network environments or devices to identify configuration issues. SSL/TLS Configuration: Issues with SSL/TLS configurations may prevent secure WebSocket connections. Solution: Verify SSL/TLS certificates, encryption protocols, and cipher suites. Ensure that both client and server support the same secure connection parameters. WebSocket Protocol Mismatch: Incompatibility between the WebSocket protocol versions used by the client and server can lead to connection failures. Solution: • Ensure consistency in the WebSocket protocol version between client and server • Confirm application compatibility with the latest WebSocket specifications.
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*If any of these issues cannot be resolved, please contact the **<National Centre>** to evaluate using the offline computer-based solution.*

PISA 2025 SRT Results Form

School Name:	
PISA School ID:	
Form completed by:	

SECTION 1: Computer environment in your school for the PISA assessment

Type of <u>IT devices</u> at your school that can be used for PISA <i>(Please tick all that apply)</i>	Type of <u>operating systems</u> at your school that can be used for PISA <i>(Please tick all that apply)</i>
Computers ☐	Windows ☐
Laptops ☐	Mac ☐
Netbooks ☐	Linux ☐
Other ☐ Please specify:	Chrome ☐
	Other ☐ Please specify:

SECTION 2: Results from running the SRT

Did your system meet the technical specifications? <i>(Please tick one response)</i>
Yes ☐
No ☐
Comments or observations